SCIENCE 7 · UNIT 4a

Waves

Exploring Wave Properties



Ms. Lena Welch · 2026–27

Science 7 · Week 1: Nov 9 – Nov 13

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · NOV 9
3.12: Unit 3 Mission (Test)

ASSIGNED **3.12 Test** (20 pts). Due Mon, Nov 16. *Extended time: Mon, Nov 23.*

SURVEY Take the **Unit 4 vocabulary pretest**. Closes Mon, Nov 16.



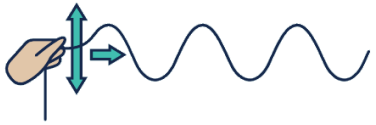
TUESDAY · NOV 10
4.01: Describe a Wave

WARM-UP Integers: **Battle of the Charges** + **Case #2** (the buoy).

READ Read lesson **4.01: Describe a Wave**.

CLASSKICK Complete the **4.01** Classkick slides: Parts 1–2.

DUE! **3.09 PhET Lab** is due today.



WEDNESDAY · NOV 11
4.02: Amplitude

WARM-UP Integers: **Case #3** (the pier post).

READ Read lesson **4.02: Amplitude**.

CLASSKICK Complete the **4.02** Classkick slides: Parts 1–2 (PhET Waves Intro).



THURSDAY · NOV 12
4.03: Frequency and Speed of a Wave

WARM-UP Integers: **Case #4** (the dolphin).

READ Read lesson **4.03: Frequency and Speed of a Wave**.

CLASSKICK Complete the **4.03** Classkick slides: Part 1, the PhET CER (Parts 2–3), and the **4.03 Quest Map** (Part 4).

ASSIGNED **4.03 Quest** (10 pts). Due Thu, Nov 19. *Extended time: Mon, Nov 30.*

DUE! **Unit 3 vocab post test** is due today.

FRIDAY · NOV 13
Science Friday

FINISH Finish any lessons, Classkick slides, or assignments you have not completed this week.

REDO Check your grades. Lower than a B? Review the Classkick notes and redo it.

CLASSKICK Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.

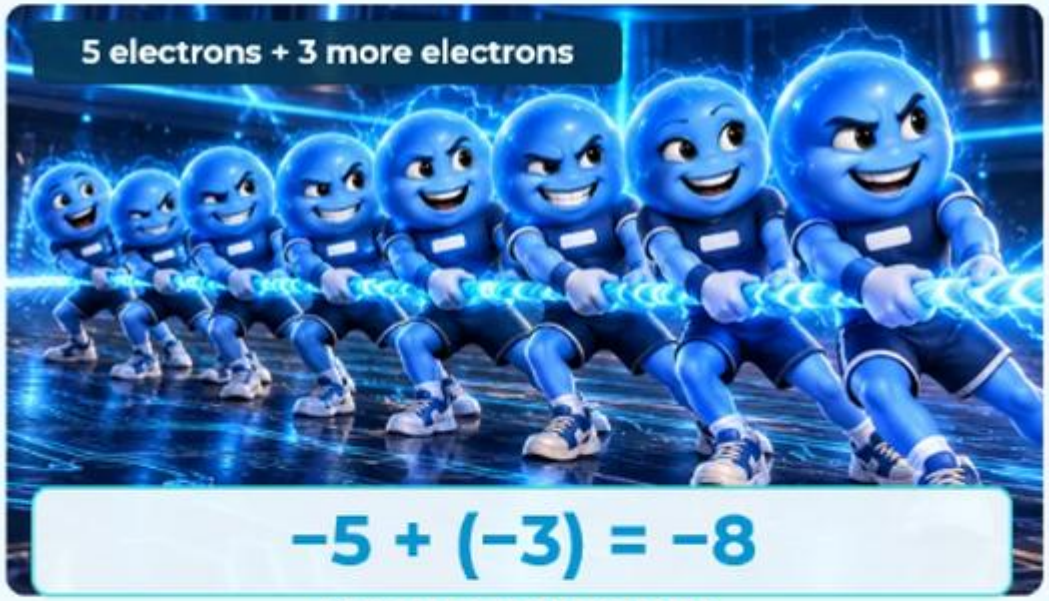
UNIT 3 Last chance to finish Unit 3 work before the 3.12 Test is due Monday, Nov 16.

BATTLE OF THE CHARGES

Adding integers with electrons and protons

SAME SIGN? TEAM UP!

Add the counts. Keep the sign.



$$-5 + (-3) = -8$$

Same signs: add and stay!

OPPOSITE SIGNS? PAIR UP!

Subtract. The sign with more wins.



$$-7 + 3 = -4$$

3 pairs total 0. Four extra electrons!

ELECTRON = −1 PROTON = +1

"More" means greater absolute value. Pairs sum to 0; particles do not disappear.

CASE #2 • TUESDAY

Is the buoy at its rest position?

WAVE WORD: *Amplitude*
The distance from the rest position to a crest or trough.

MYSTERY

A buoy is 2 m below the rest position. It moves up 2 m. Is the buoy at its rest position?

CLUES

- Start: −2 m
- Move up: +2 m
- Rest: 0 m

BUOY HEIGHT (m)

+3
+2 ● Crest +2 m
+1
0 ● Rest 0 m
−1
−2 ● Trough −2 m
−3



CLAIM

Answer the mystery question.

Is the buoy at its rest position?

☐ YES ☐ NO

EVIDENCE

Show your work!

Solve:

−2 + 2 = _____

REASONING

Explain the rule.

Finish the sentence:

"−2 and 2 are opposites, so..."

4.01: Describe a Wave

Part 1: What Is a Wave?

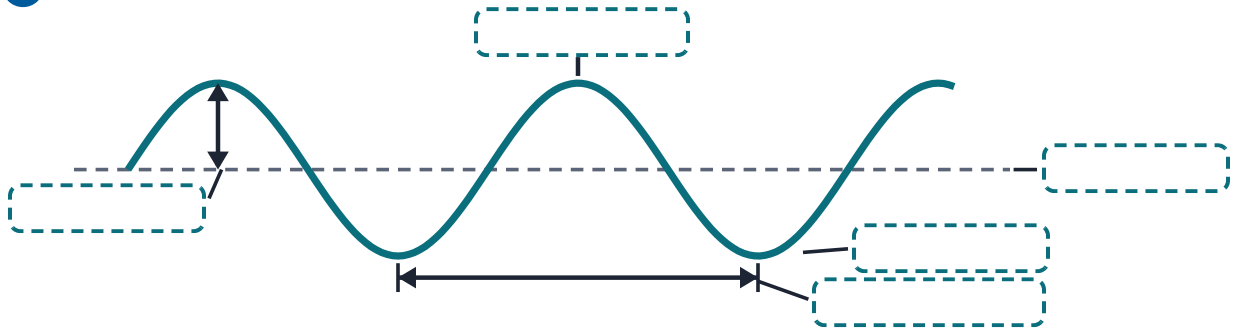
Interactive Link
Label the Parts of a Wave (Wordwall)

★ Key Ideas

- A **wave** is a disturbance that carries **energy** from one place to another. **Vibrations** cause waves.
- **Mechanical waves** need a **medium** (water, rope, air) to travel through. The medium moves, then returns to its **rest position**.
- **Transverse**: the medium moves up and down (*across* the wave).
- **Longitudinal:** the medium moves back and forth (*along* the wave).
- **Crest** = high point. **Trough** = low point.

boxed = Unit 4 focus word

1 Drag a word into each box to label the wave. DRAG USE THE LINK ON THIS SLIDE



Word bank: wavelength | trough | frequency | crest | rest position | amplitude *One extra word.*

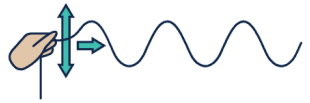
2 Drag a word into each blank. DRAG

Waves are caused by _____. A mechanical wave carries _____ from one place to another. It needs a _____ to travel through. The medium moves, then goes back to its _____ position.

Word bank: medium | speed | rest | vibrations | energy *One extra word.*

3 Circle the type of wave. CIRCLE

Shaking a rope



medium moves up ↕ down
energy moves →

transverse | longitudinal

Pushing a spring toy



medium moves back ↔ forth
energy moves →

transverse | longitudinal

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.01: Describe a Wave

Part 2: Measuring Waves

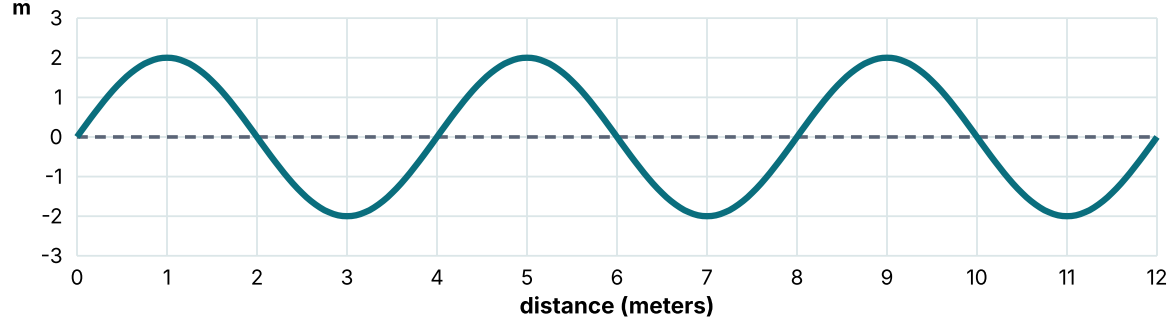
Interactive Link
Practice Measuring Wavelengths (PBS)

★ Key Ideas

- **Cycle:** one complete **wave** that repeats (one crest + one trough). A **waveform** is a curve showing the shape of a wave.
- **Wavelength:** the length of one cycle. Measure crest to crest *or* trough to trough. You get the same answer both ways!
- **Amplitude:** the distance from the rest position to a crest (or to a trough). **Bigger amplitude = more energy.**
- **Frequency:** how many cycles pass in a period of time. More cycles in the same time = higher frequency.
- **Speed:** how far the wave travels each second. **speed = wavelength × frequency**

boxed = Unit 4 focus word

5 Measure the wavelength of this rope wave **two ways**, then the amplitude. Type your answers in the boxes. FILL IN



Wavelength, crest to crest: [ ] m Trough to trough: [ ] m Same both ways? YES NO Amplitude: [ ] m

6 Both waves pass by in 1 second. Circle the wave with the **higher frequency**. How many cycles does it have? CIRCLE

Remember: one cycle = one full up and one full down.



Higher frequency: Wave A Wave B

It has [ ] cycles.

7 Open the link. Click **Measurement**. Drag the ruler to measure each wave. Record your data, then click Check Answers. FILL IN

USE THE PBS LINK

Wave	Wavelength
Top wave	[ ] units
Middle wave	[ ] units
Bottom wave	[ ] units

Click **Light Spectrum**. Red light has a longer shorter wavelength than violet light.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #3 • WEDNESDAY

Is Crest B past the pier post?

WAVE WORD: *Crest*
The highest point of a wave.

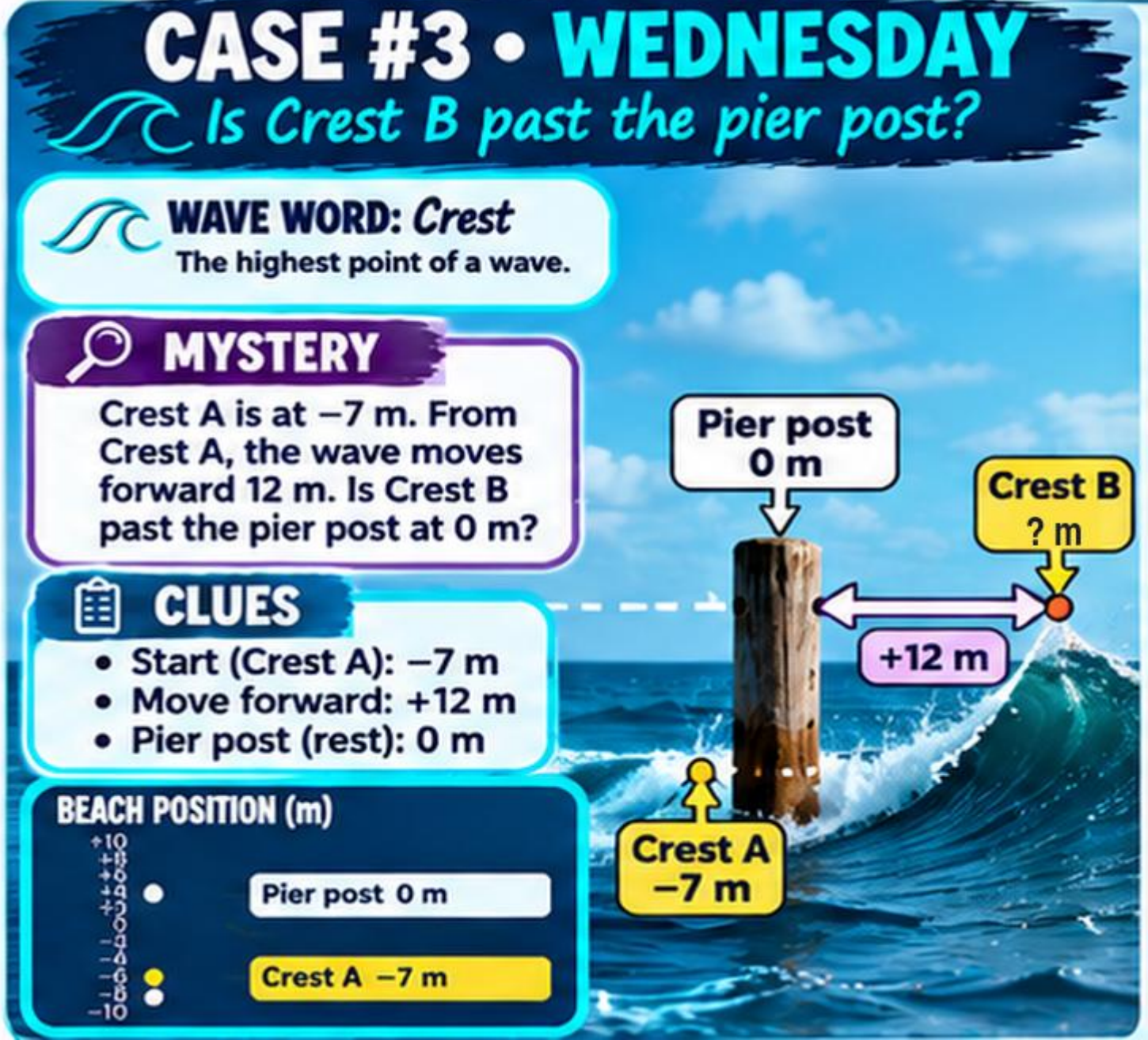
MYSTERY

Crest A is at −7 m. From Crest A, the wave moves forward 12 m. Is Crest B past the pier post at 0 m?

CLUES

- Start (Crest A): −7 m
- Move forward: +12 m
- Pier post (rest): 0 m

BEACH POSITION (m)



CLAIM

Answer the mystery question.

Is Crest B past the pier post?

☐ YES ☐ NO

EVIDENCE

Show your work!

Solve:

−7 + 12 = _____

REASONING

Explain the rule.

Finish the sentence:

"12 has the larger absolute value, so the answer is..."

4.02: Amplitude

How does amplitude affect the energy of a wave?

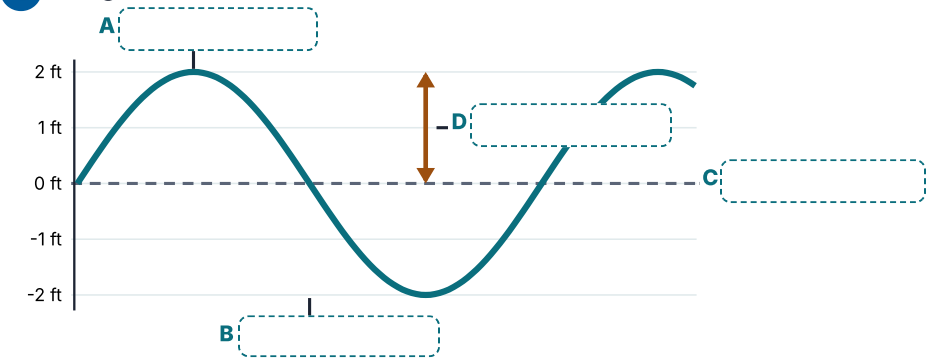
Interactive Link
PBS Amplitude Demonstration: click "amplitude," then move the blue slider

★ Key Ideas

- **Amplitude** = how far a **wave** moves from its resting position to a crest or trough.
- **Energy** = the ability to do work or cause change. **Bigger amplitude = more energy.**
- Louder sounds and bigger ocean waves have larger amplitudes.

boxed = Unit 4 focus word

1 Drag a word to label each box on the wave. DRAG



Word bank: trough | amplitude | wavelength | crest | resting position *One extra word.*

2 Use the scale. Drag: the amplitude of the wave above is _____ feet. 2 | 4 DRAG

3 Circle the wave that carries MORE energy. CIRCLE



4 Circle the correct word in each sentence. CIRCLE
USE THE LINK ON THIS SLIDE

a) Moving the slider to **high** amplitude makes the wave taller or flatter

b) Plucking a guitar string **harder** makes a louder or quieter sound.

c) Bigger ocean waves cause more or less erosion on beaches.

d) Stronger earthquakes make waves with larger or smaller amplitudes.

5 How are amplitude and energy related? Drag a word into each blank. DRAG

A wave with a _____ amplitude carries _____ energy.

bigger | more | less *One extra word.*

6 What happens to the amplitude of sound waves when you turn UP the volume? Why? Drag a word into each blank. DRAG

The amplitude gets _____. A louder sound carries _____ energy.

more | smaller | bigger *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.02: Amplitude

Part 2: Exploring Amplitude

Interactive Link 1
Waves Intro (PhET)

Interactive Link 2
Simple Wave Simulator (Physics Classroom)

★ Key Ideas

- **Amplitude**: how far the **medium** moves from its rest position.
- Bigger amplitude = **more energy**. This is a direct relationship: when amplitude goes up, energy goes up too.
- For sound, bigger amplitude = **louder** (pluck a guitar string harder). Bigger ocean waves cause more erosion.
- Changing amplitude does **NOT** change the **wavelength** or the **frequency**.

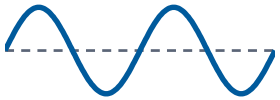
boxed = Unit 4 focus word

7 Link 1: **Water** screen. Turn on the water. Move the **Amplitude** slider low, then high. Circle what you see. CIRCLE USE THE LINK ON THIS SLIDE

low amplitude



high amplitude



a) High amplitude makes (taller) (shorter) waves.

b) The drops hit the water (harder) (softer).

c) The waves carry (more energy) (less energy).

8 Link 1: **Sound** screen. Turn the sound ON. Move the **Amplitude** slider up. Circle what changes. CIRCLE USE THE LINK ON THIS SLIDE

a) The sound gets (louder) (quieter)

b) The speaker moves (farther) (less) back and forth

c) The pitch (frequency) (goes up) (stays the same)

d) Turning up the volume makes the amplitude (bigger) (smaller)

9 Link 2: Simple Wave Simulator. Change **only** the amplitude. What happens to each part? CIRCLE USE THE LINK ON THIS SLIDE

Part of the wave	When amplitude goes up, it...
Height of the wave	(changes) (stays the same)
Wavelength (crest to crest)	(changes) (stays the same)
Number of waves on the screen	(changes) (stays the same)

10 Sum it up. Drag a word into each blank. DRAG

Amplitude is how far the medium moves from its _____. A wave with a bigger amplitude carries more _____. A sound with a bigger amplitude is _____. Changing the amplitude does not change the _____.

(wavelength) (speed) (energy) (rest position) (louder) *One extra word.*

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CASE #4 • THURSDAY

Is the dolphin at its pod's depth?

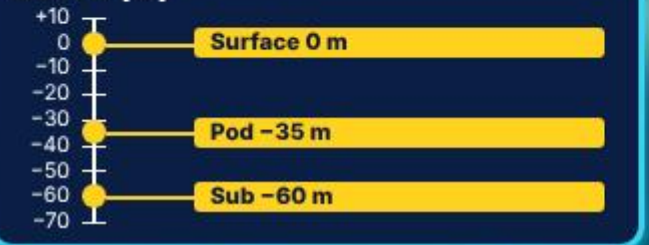
WAVE WORD: *Depth*
The distance below the surface.

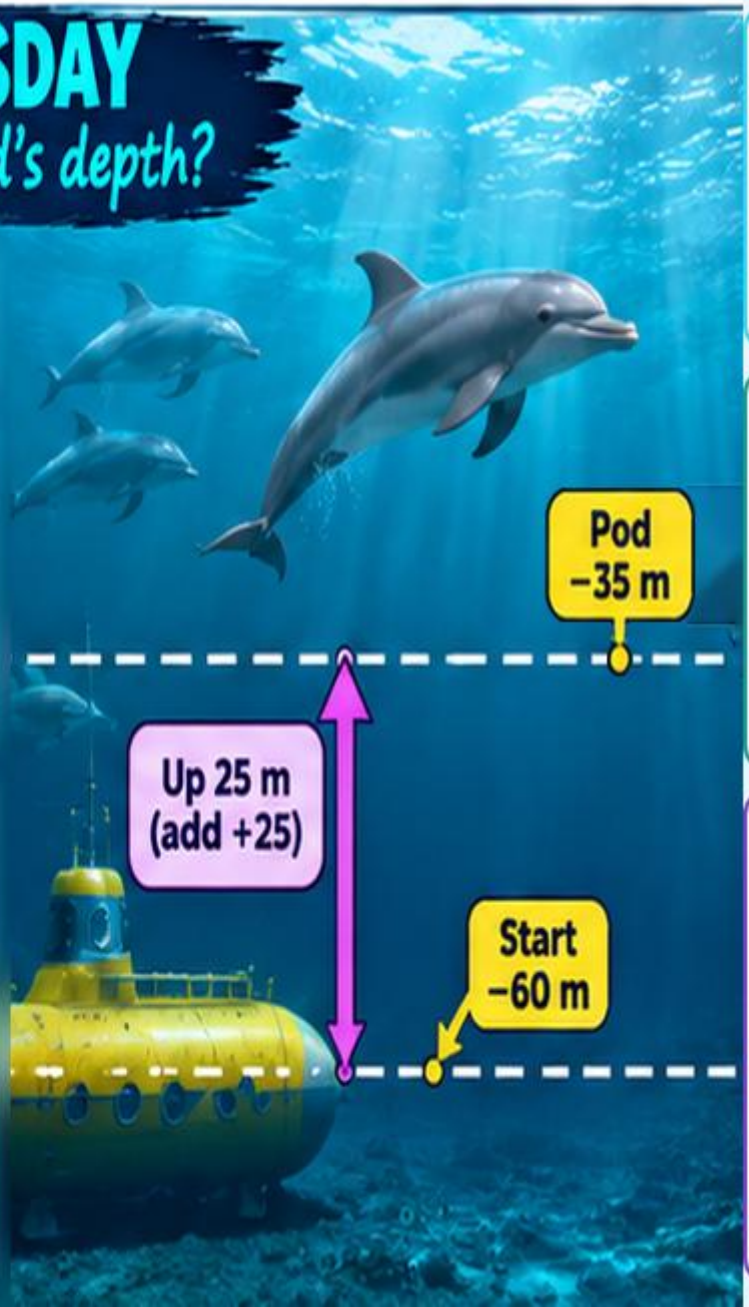
MYSTERY

A dolphin swims by a submarine 60 meters below the surface (−60 m). It swims up 25 m. Its pod swims 35 meters below the surface (−35 m). Is the dolphin at its pod's depth?

CLUES

- Start (by the sub): −60 m
- Move up: +25 m
- Pod's depth: −35 m





CLAIM

Is the dolphin at its pod's depth?

☐ YES ☐ NO

EVIDENCE

Solve:

−60 + 25 = _____

The dolphin is now at _____ m.

REASONING

Finish the sentence:

The signs are different, so I subtract
60 − 25 = _____ and keep the sign of _____.
The dolphin is at _____ m, the _____ as its pod.

4.03: Wave Speed and Frequency

Part 1: Frequency and Wavelength

Interactive Link 1
Exploring Frequency (PBS)

Interactive Link 2
Frequency and Wavelength (PBS)

★ Key Ideas

- **Frequency**: the number of waves that go past a point in a given time.
- Frequency is measured in **hertz (Hz)**. 1 Hz = 1 **wave** per second.
- **Wavelength**: the distance from one crest to the next crest.
- **High frequency** waves have **short** wavelengths (waves close together).
- **Low frequency** waves have **long** wavelengths (waves spread out).
- Changing the **amplitude** does **NOT** change the wavelength.

boxed = Unit 4 focus word

1 Open Link 1. Read with the next and back buttons, then move the frequency slider. Circle your answers. CIRCLE

USE LINK 1: EXPLORING FREQUENCY

a) Raise the frequency. The waves get
closer together or farther apart

b) Raise the frequency. Each second,
more or fewer waves pass a point.

c) Frequency is measured in
hertz (Hz) or meters (m)

2 Both waves were drawn over 1 second. Count the waves, then drag a number into each box. DRAG



Frequency of A: ____ Hz Frequency of B: ____ Hz

Which wave has the higher frequency? A B

Which wave has the shorter wavelength? A B

5 2 4 *One extra word.*

3 Drag a word into each blank. DRAG

Frequency is the _____ of waves that pass a point each second. It is measured in _____. Wavelength is the _____ from one crest to the next crest. High frequency waves have _____ wavelengths. Low frequency waves have _____ wavelengths.

distance short number speed long hertz *One extra word.*

4 Move the slider on Link 2. Then drag a word into each blank. DRAG

USE LINK 2

When the frequency is _____, the waves are close together, so the wavelength is _____. When the frequency is _____, the waves are spread out, so the wavelength is _____.

fast high short low long
One extra word.

5 A high whistle and a low drum sound at the same time across the same room. Circle your answers. CIRCLE

a) Which reaches you first?
the high whistle the low drum same time

b) Wave speed depends on the
frequency medium

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.03: Wave Speed and Frequency

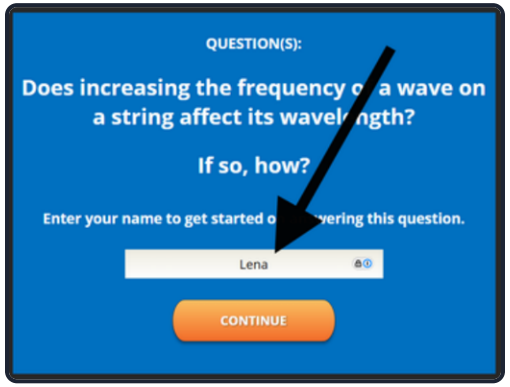
Part 2: CER Lab Steps

Interactive Link
Frequency and Wavelength CER (Virtual Science Teachers)

CER Lab: Frequency and Wavelength (Virtual Science Teachers)

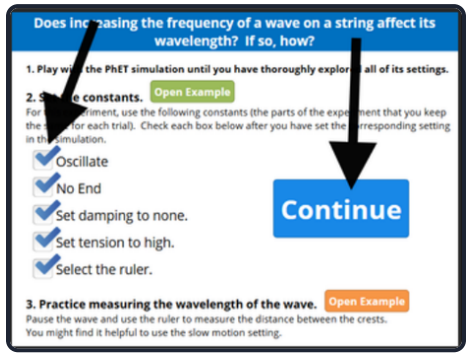
USE THE LINK ON THIS SLIDE

Step 1. Type your name in the box on the website. Then click **CONTINUE**.



Step 2. The PhET lab is at the **top** of the page. The directions are at the **bottom**.
Step 3. Scroll to the top and set up the lab: damping **none**, tension **high**, **No End**, **Oscillate**, and select the **ruler**. Click **Open Example** on the website if you need help.

Step 4. Check off each setting after you set it. Practice measuring: pause the wave and use the ruler to measure from **crest to crest**. Then click **Continue**.



> Our question: **"Does increasing the frequency of a wave on a string affect its wavelength? If so, how?"**

Step 5. In the PhET lab, set the frequency to **1.0 Hz**. Use the ruler to measure the wavelength. Record it in the data table at the bottom of the website.
Step 6. Repeat for **1.5, 2.0, 2.5,** and **3.0 Hz**.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

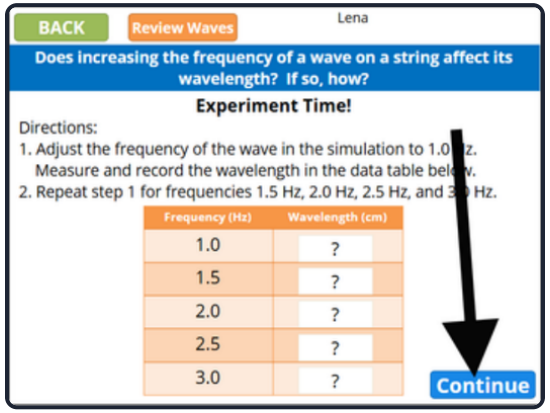
4.03: Wave Speed and Frequency

Part 3: Write Your CER

Interactive Link
Frequency and Wavelength CER (Virtual Science Teachers)

CER Lab continued: Write your CER

Step 7. When you have entered all your data, click **CONTINUE**.



Step 8. Use **YOUR DATA** to fill in the **Claim**, **Evidence**, and **Reasoning** boxes on the website.

Step 9. Click the blue **?** to see sentence starters and examples for each part. Use them to help you write.

Step 10. When you are done, take a **screenshot** of your CER and paste it in the box on the right.

You will receive points as shown in the rubric.

Wave Frequency & Wavelength Lab – Points Rubric

Not Met 0–2	Data table mostly incomplete or incorrect. Few frequencies recorded OR major errors. Written C-E-R sentences (if present) are incomplete or unclear. If either the data table or written portion is blank, score stays in this range.
OK 3–4	Most or all data recorded with some inaccuracies. C-E-R sentences partly complete but may be choppy or missing detail. Shows partial understanding.
Great 5–6	Data table fully completed with accurate or reasonable measurements. C-E-R written in mostly complete sentences. Shows correct understanding (wavelength decreases as frequency increases). Full credit only if BOTH the data table and written portion are complete.

📸 **Paste your CER screenshot here**

Claim · Evidence · Reasoning

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.03: Wave Speed and Frequency

Part 4: Quest Map (review for the 4.03 Quest)

Drag a word from the Answer Bank into each box. One extra word.

Need help? Look back at the Classkick slide for the lesson in each part's title. The questions here match the 4.03 Quest. Want more practice? Try the **Waves Review games** on the 4.03 lesson page.

Answer Bank *One extra word.*

longer | shorter | increases | energy | same time | 5 m | resting position | decreases | -3 m | crests | Wave 4

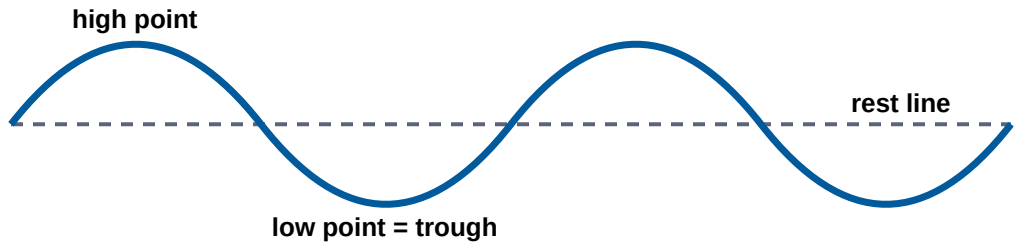
Part 1: Parts of a Wave Lesson 4.01



A student shakes a rope tied to a tree. A wave moves down the rope.

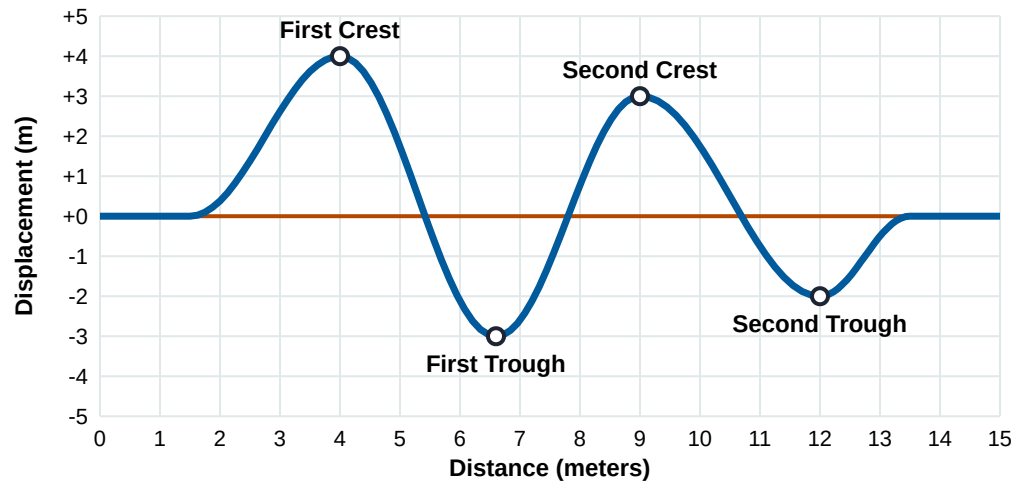
1. The high points of the wave on the rope are called _____.

2. After the student stops and the energy passes through, the rope particles return to their _____.



Don't fall for it: Troughs are the LOW points. The rope does not travel to the tree, and it does not stay stretched at the high points. Only the **energy** moves along.

Part 2: Reading a Wave Graph Lesson 4.01



3. The wavelength from the first crest (4 m) to the second crest (9 m) is _____.

4. At its lowest point, the first trough reaches a displacement of _____.

Don't fall for it: Wavelength is the distance **between** crests (9 − 4), not the height of a crest. Troughs go **below** zero, so their numbers are negative.

Part 3: Amplitude and Energy Lesson 4.02

A class measures ocean waves hitting a seawall.

Wave	Amplitude	Energy
Wave 1	2 ft	10 J
Wave 2	4 ft	40 J
Wave 3	6 ft	90 J
Wave 4	8 ft	160 J

5. As the amplitude increases, the energy _____.

6. _____ is most likely to erode the beach, because its large amplitude carries the most _____.

Don't fall for it: Amplitude does not change a wave's frequency or speed. Bigger amplitude = more **energy**.

Part 4: Frequency, Wavelength, and Speed Lesson 4.03



7. When the frequency increases, the waves squeeze closer together, so the wavelength _____.

8. Wave A has a high frequency. Compared to Wave B, Wave A has a _____ wavelength.

9. A high-pitch and a low-pitch sound travel across the same classroom. They arrive at the _____.

Don't fall for it: Frequency does **not** change speed. In the same medium, all sound waves travel at the same speed.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

Friday is a school day! Spend at least 1 hour on science today.

Science Friday! Week 1 · Friday, Nov 13

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	GRADED **3.12 Test** (Unit 3 Mission) · 20 pts	**Closes Mon, Nov 16** ext: Mon, Nov 23	Yes No
☐	SURVEY Unit 4 vocabulary pretest	**Closes Mon, Nov 16**	Done = check it!
☐	CLASSKICK Integer Battle + Case #2, 4.01 slides (4)	With 4.14 Test*	Yes No
☐	CLASSKICK Case #3, 4.02 slides (3)	With 4.14 Test*	Yes No
☐	CLASSKICK Case #4, 4.03 slides + Quest Map	With 4.14 Test*	Yes No
☐	GRADED **4.03 Quest** · 10 pts	**Closes Thu, Nov 19** ext: Mon, Nov 30	Yes No

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished

Coming Up Next Week

Nov 16 – Nov 20 · Red tags show what **closes** that day.

MON 16
4.04 Wave Speed and Density
CLOSES: 3.12 TEST
CLOSES: UNIT 4 VOCAB PRETEST
DUE: Unit 3 Classkick packet
DUE: 4.04 Refraction Lab · due Nov 23

TUE 17
4.05 Waves and Their Medium

WED 18
4.06 Transverse Waves

THU 19
4.07 Longitudinal Waves
CLOSES: 4.03 QUEST

FRI 20
Science Friday!



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Science 7 · Week 2: Nov 16 – Nov 20

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · NOV 16
4.04: Wave Speed and Density

WARM-UP Integers: **Flip the Sign** + **Case #1** (the gym).

READ Read lesson **4.04: Wave Speed and Density**.

CLASSKICK Complete the **4.04** Classkick slides: Part 1 Refraction and the Refraction Lab.

ASSIGNED **4.04 Refraction Lab** (10 pts). Due Mon, Nov 23. *Extended time: Mon, Nov 30.*

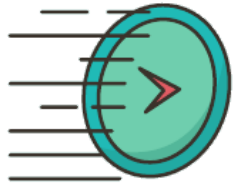
DUE! **3.12 Test**, **Unit 3 Classkick packet**, and **Unit 4 vocab pretest** are due today.

TUESDAY · NOV 17
4.05: Waves and Their Medium

WARM-UP Integers: **Case #2** (the rubber duck).

READ Read lesson **4.05: Waves and Their Medium**.

CLASSKICK Complete the **4.05** Classkick slides: your Wave Speed and Temperature Lab (Parts 1–2) and the water graph (Part 3).

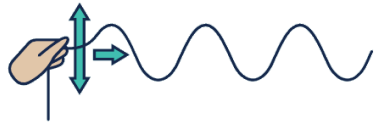


WEDNESDAY · NOV 18
4.06: Transverse Waves

WARM-UP Integers: **Case #3** (the slinky).

READ Read lesson **4.06: Transverse Waves**.

CLASSKICK Complete the **4.06** Classkick slides: Parts 1–2.



THURSDAY · NOV 19
4.07: Longitudinal Waves

READ Read lesson **4.07: Longitudinal Waves**.

CLASSKICK Complete the **4.07** Classkick slides: Parts 1–3 and the **4.07 Quest Map**.

ASSIGNED **4.07 Quest** (10 pts). Due Mon, Nov 30. *Extended time: Thu, Dec 3.*

DUE! **4.03 Quest** is due today.

FRIDAY · NOV 20
Science Friday

FINISH Finish any lessons, Classkick slides, or assignments you have not completed this week.

REDO Check your grades. Lower than a B? Review the Classkick notes and redo it.

CLASSKICK Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.

BREAK Thanksgiving Break is Wed–Fri next week. Plan to finish your work on Mon and Tue.

MISSION: FLIP THE SIGN!

Subtracting integers • Add the opposite

The Integer Crew is trapped! Recode subtraction as addition to unlock the gate.

1. **KEEP** the first.
2. **CHANGE** to +.
3. **FLIP** the second!

SUBTRACT A POSITIVE

The second number flips to negative.



$7 - 10 = 7 + (-10) = -3$

SUBTRACT A NEGATIVE

The second number flips to positive.



$-9 - (-2) = -9 + 2 = -7$

FIND THE CHANGE: end − start → $6 - 11 = -5$

Negative change = down or left.

A math model: robots represent signed numbers.

CASE #1 • MONDAY

How much warmer is the gym?

WAVE WORD: ***Wave speed***
How fast a wave travels through a medium.

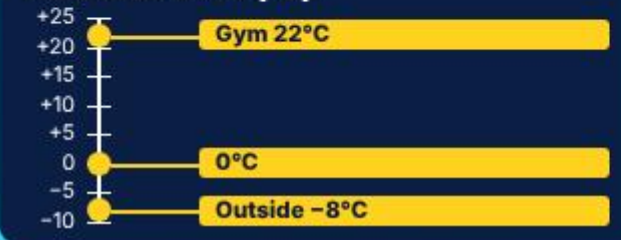
MYSTERY

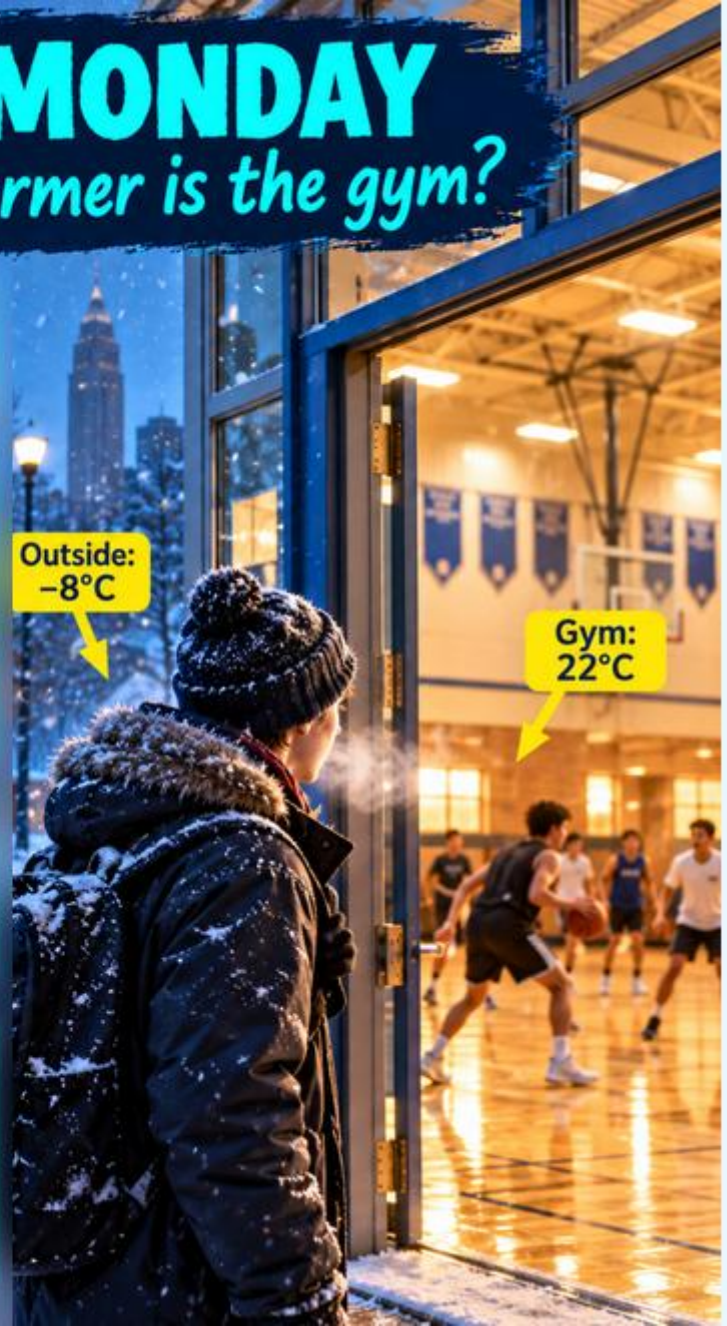
Sound waves travel faster in warm air than in cold, denser air. On a January morning in Cleveland, it is 8 degrees below zero (Celsius) outside. Inside the school gym, it is 22°C. A student says the gym is only 14 degrees warmer. Is the student right?

CLUES

- Outside: −8°C
- Gym: 22°C
- Student says: 14°

THERMOMETER (°C)





CLAIM

Is the student right?

☐ **YES** ☐ **NO**

EVIDENCE

Solve:

22 − (−8) = _____

Rewrite as adding: 22 + _____ = _____

REASONING

Finish the sentence:

22 − (−8) is the same as

22 + _____ = _____.

So the gym is _____ degrees warmer.

4.04: Wave Speed and Density

Part 1: Refraction

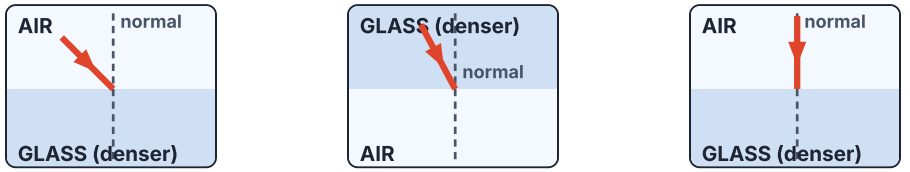
Interactive Link
Refraction Video (4.04 page)

★ Key Ideas

- **Refraction**: the bending of light (or any **wave**) when it enters a different **medium**.
- **Density**: how much mass is packed into a space. Glass and water are denser than air.
- In a new medium the **speed changes**. The **frequency stays the same**, so the **wavelength changes**.
- Into a **denser** medium: light slows down and bends **toward the normal**.
- Into a **less dense** medium: light speeds up and bends **away from the normal**.
- Straight in (at a right angle): **no bending**.
- **Normal:** an imaginary line at a right angle to the surface.

boxed = Unit 4 focus word

1 Watch the video. A light ray (red) hits a new medium. Circle what the ray does next. CIRCLE USE THE LINK ON THIS SLIDE



a) Air into glass	b) Glass into air	c) Straight in
toward	toward	toward
away	away	away
straight	straight	straight

toward / away = from the normal (dashed line)

2 The marching band walks onto a hill at an angle. Drag a word into each blank. DRAG



The marchers who reach the hill first _____ down. To stay in rows, the line _____ direction. Light does the same thing when it enters a _____ medium.

denser | faster | slow | changes — *One extra word.*

3 Light goes from air into glass. Circle what happens to each part of the wave. CIRCLE

Speed	speeds up	slows down	stays the same
Frequency	changes	stays the same	
Wavelength	changes	stays the same	
Direction	bends	goes straight	(the ray hits at an angle)

Our eyes use refraction to **focus light** or **make light** on the back of the eye.

4 A light ray goes straight down into a pool, at a right angle to the water. Drag a word into each blank. DRAG

The ray does _____ because it hits the water straight on. Its speed still changes: it _____ because water is _____ than air.

not bend | denser | speeds up | slows down
One extra word.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

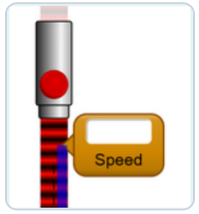
4.04: Wave Speed and Density

Lab: Refraction Lab (Bending Light)

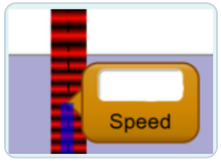
Interactive Link
Bending Light: More Tools (PhET)

Lab Steps

- Open the link. It opens **More Tools**.
- Switch from **Ray** to **Wave**. Set the wavelength to **650 nm**.
- Press the red button to turn on the laser. Leave the laser at its starting angle.
- Use the material arrows on the right. Set the top to **AIR** and the bottom to **AIR**.
- Drag the **Speed** tool out. Put it on the beam at the top, then on the beam at the bottom.
- Record the speeds. Then change the bottom to **GLASS**, then **WATER**.



Speed in the top



Speed in the bottom

The speed tool shows speed compared to **c**, the speed of light in empty space.

5 Record your data. FILL IN USE THE LINK ON THIS SLIDE

Top	Bottom	Does the beam bend?	Speed top	Speed bottom
Air	Air			
Air	Glass			
Air	Water			

6 Use your data. Circle your answers. CIRCLE

a) Light traveled slowest in **air** **water** **glass**

b) Air to air, the beam **went straight** **bent**

c) When the light slowed down, the beam bent **toward the normal** **away from the normal**

★ Key Ideas

- Light slows down when it goes from air into **water** or **glass**.
- When light slows down at an angle, it bends **toward the normal**.
- Same **medium** on both sides = **no change** in speed or direction.
- **Watch out:** light is *slower* in denser materials. Sound is the opposite: it is *faster* in solids.

boxed = Unit 4 focus word

7 When light goes from a less dense medium into a more dense medium, what happens to its speed? Why does the beam bend? Drag a word into each blank. DRAG

The light _____, from 1.00 c in air to _____ in glass. One side of the beam slows first, so the beam _____ toward the _____.

speeds up **0.67 c** **slows down** **bends** **normal** *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.04 Refraction Lab Map

Notes for the 4.04 Refraction Lab (graded)

Drag a word from the Answer Bank into each box. One extra word.

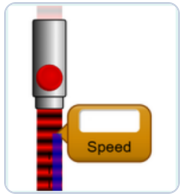
Need help? Look back at the Classkick slide for the lesson in each part's title. The questions here match the 4.04 Refraction Lab. Test each answer in **PhET Bending Light (More Tools)**.

Answer Bank *One extra word.*

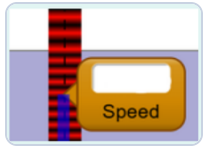
slows down | stays the same | bend | straight | speeds up | bends down | dense

Part 1: Set Up the Lab Lesson 4.04

1. Click the PhET Lab link. Select **More Tools**.
2. Top left: select **Wave** and set it to **650 nm**. Press the **red button**.
3. Right side: click the **material arrows** to pick the top and bottom materials.
4. Left side: drag out the **Speed** tool and put it on the beam.



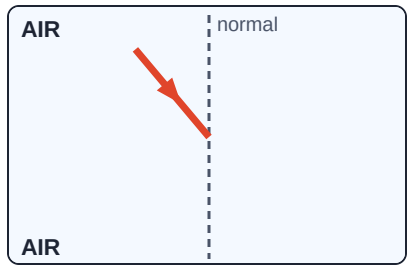
Speed tool on the top beam



Speed tool on the bottom beam

Tip: Look at the beam from the top of the laser. The dashed line is the **normal**.

Part 2: Air → Air Questions 1 and 2

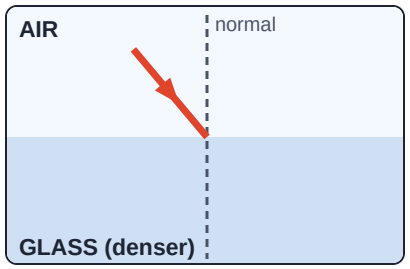


1. With AIR on top and AIR on the bottom, the wave goes _____.

2. The speed tool shows the ray's speed _____.

Don't fall for it: Same medium on both sides = **no** change in speed and **no** bending.

Part 3: Air → Glass Questions 3 and 4

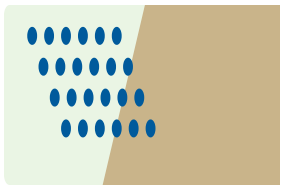


3. With AIR on top and GLASS on the bottom, the ray _____ (toward the normal).

4. The speed tool shows the ray _____ in the glass.

Don't fall for it: Light does **not** speed up in glass. Glass is denser than air, so light goes **slower**.

Part 4: Why Does It Bend? Question 5



Like a marching band walking onto a hill: the side that reaches the slow part first slows down, so the whole line turns.

5. When light goes from a less dense medium into a more dense medium, the light will slow down and _____ because the new medium is more _____.

Don't fall for it: The light does not speed up, and it does not stay the same. Denser medium = **slower** light = **bending**.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #2 • TUESDAY

Did the rubber duck drop 5 meters?

WAVE WORD: *Medium*
The material a wave travels through.

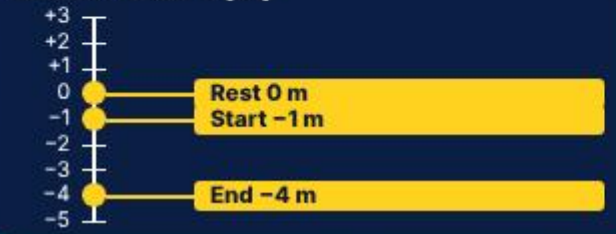
MYSTERY

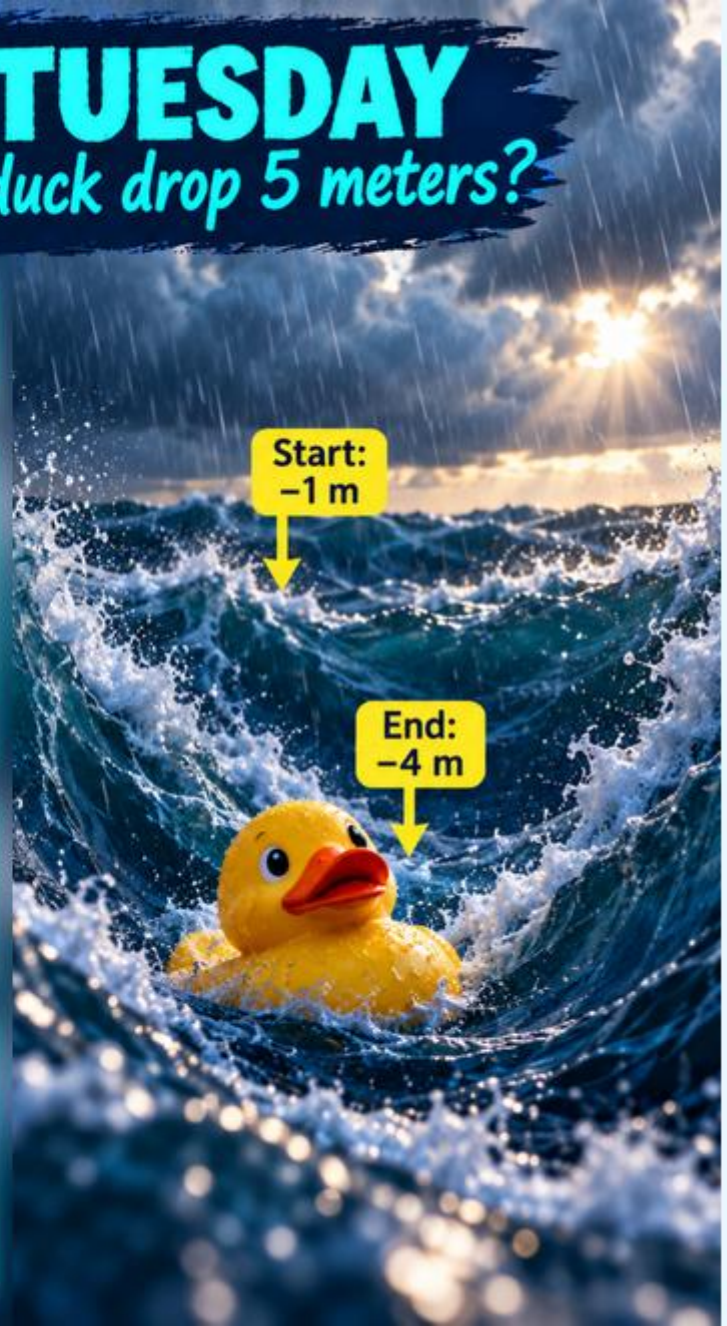
A rubber duck floats on stormy Lake Erie. The water is the medium, so the duck bobs up and down as waves pass instead of traveling with them. The duck sits 1 meter below rest position. Then a deep trough drops it to 4 meters below rest position. Did the duck drop 5 meters?

CLUES

- Start: −1 m
- End: −4 m
- Change = end − start

DUCK HEIGHT (m)





CLAIM

Did the duck drop 5 meters?

☐ YES ☐ NO

EVIDENCE

Solve:

−4 − (−1) = _____

Rewrite as adding: −4 + _____ = _____

REASONING

Finish the sentence:

−4 − (−1) is the same as

−4 + _____ = _____.

So the duck dropped _____ m.

4.05: Waves and Their Medium

Part 1: Counting Thunder (Lab Parts 1–2)

Interactive Link 1
Speed of Sound and Temperature Video

Interactive Link 2
Wave Speed and Temperature Lab

★ Key Ideas

- Lightning makes **light** and **sound** at the same time.
- **Light** (EM **wave**) is about **300,000,000 m/s**. It reaches you almost instantly.
- **Sound** (mechanical wave) is about **343 m/s** in air. Air particles bump each other to pass it along.
- **speed = distance ÷ time**
- The **medium** (solid, liquid, or gas) changes how fast a wave travels.
- Sound is **fastest in solids**, slower in liquids, and **slowest in gases**.
- **Density**: the amount of mass in a given volume.

boxed = Unit 4 focus word

Lab Steps: Parts 1 and 2

START WITH LINK 1: THE VIDEO

Watch the video (Link 1). Then open the **Wave Speed and Temperature Lab** (Link 2).

- **Part 1:** Pick 1, 5, or 9 miles. Click **Make lightning**. Click **I hear thunder** right when you hear the BOOM.
- Do each distance **3 times**. The Top table fills in for you. Copy it onto this slide.
- **Part 2: Why it works.** Click **Make lightning** and watch light and sound race. Read the boxes.

1 Part 1: Copy your Top table. FILL IN USE LINK 2: THE LAB

Position (miles)	Distance (m)	Counted seconds Trial 1	Counted seconds Trial 2	Counted seconds Trial 3	Average Counted Seconds	Calculated Speed (m/s)
1	1,609 m					
5	8,047 m					
9	14,484 m					

Your counts will not be exact. That is OK!

2 Part 2: Why it works. Circle your answers. CIRCLE

a) Which gets to you first? **light** **sound**

b) Light is an **electromagnetic** **mechanical** wave. It **needs** **does not need** air.

c) Sound is a(n) **electromagnetic** **mechanical** wave. It needs a **medium** **vacuum**

d) Speed of sound in air: about **343 m/s** **300,000,000 m/s**

3 Part 2: How to use it in a real storm. FILL IN

Sound goes about 1 mile every **5 seconds**.

You count **15 seconds**: 15 ÷ 5 = ____ miles away.

You count **10 seconds**: ____ ÷ 5 = ____ miles away.

Stay safe: When thunder roars, go indoors. Wait 30 minutes after the last thunder.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.05: Waves and Their Medium

Part 2: Temperature (Lab Parts 3–5)

Interactive Link
Wave Speed and Temperature Lab

★ Key Ideas

- Sound travels **faster in warm air**, about 0.6 m/s faster for each 1 °C.
- Hot air particles move faster, so they **bump and pass the sound along faster**.
- Distance changes the **time**, not the **speed**.
- Changing the **medium** (like its temperature) **changes the wave speed**.
- **Elasticity:** how well a material springs back. Stiff materials like steel pass waves fast.
- At a new medium a wave can **refract** (bend), **reflect** (bounce back), or be **absorbed** (taken in).

boxed = Unit 4 focus word

4 Part 3: Temperature lab. Copy your Bottom table. FILL IN USE THE LAB LINK

Do one distance at a time (1, 5, then 9 miles). Click each temperature, then click **Save**.

Distance (miles)	Expected Travel Time @ 20 °F (s)	Speed of Sound @ 20 °F (m/s)	Expected Travel Time @ 60 °F (s)	Speed of Sound @ 60 °F (m/s)	Expected Travel Time @ 100 °F (s)	Speed of Sound @ 100 °F (m/s)
1						
5						
9						

Think: Look across a row. At the same distance, which temperature made the thunder arrive soonest? A shorter time means a faster speed.

5 Part 4: Start the race. Which gets to the end first? CIRCLE

cold air (20 °F) hot air (100 °F)

Hot air dots move faster slower than cold air dots.

So in hot air, sound passes from particle to particle faster slower

6 Part 5: Questions 1, 4, and 5. Circle your answers. CIRCLE

1. Which is faster? a light wave a sound wave

4. Sound travels faster in warm air cold air

5. If you only change the medium (the air), the wave speed changes stays the same

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.05: Waves and Their Medium

Part 3: Graph: Speed of Sound in Water

Interactive Link
Speed of Sound in Water graph (Virtual Science Teachers)

★ Key Ideas

- Sound travels **faster in warmer** water, just like in warmer air.
- Warmer particles move faster, so they pass the vibration along faster.
- Sound in water (about 1,480 m/s) is about **4 times faster** than in air (about 343 m/s).
- A **graph** shows the **trend** (pattern) at a glance.

boxed = Unit 4 focus word

7 Open the **Speed of Sound in Water** graph. Click through the reveal. Circle your answers. CIRCLE USE THE LINK ON THIS SLIDE

a) The x-axis shows
water temperature (°C) **speed (m/s)**

b) The y-axis shows
water temperature (°C) **speed of sound (m/s)**

c) As the water gets warmer, the line goes
up **down** **stays flat**

8 Read the graph. Drag the speed for each temperature. DRAG

Water temperature	Speed of sound (m/s)
0 °C	about _____
10 °C	about _____
20 °C	about _____
30 °C	about _____
40 °C	about _____

1,450 **1,350** **1,480** **1,400** **1,530** **1,510** *One extra word.*

9 Compare. Circle your answers. CIRCLE

a) In warmer water, sound travels **faster** **slower**

b) This **matches** **does not match** what you found for air in the lab.

c) At 20 °C, sound is faster in **air** **water**

d) The speed goes up by about the same amount every 10 °C. **true** **false: it goes up less each time**

10 What does the graph show that is harder to see in a data table? Drag a word into each blank. DRAG

The graph shows the _____ at a glance. The line goes _____, so sound gets _____ as water gets warmer.

faster **trend** **up** **down** *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #3 • WEDNESDAY

Did the coil move 8 cm to the left?

WAVE WORD: ***Longitudinal***

The medium moves back and forth in the same direction the wave travels.

MYSTERY

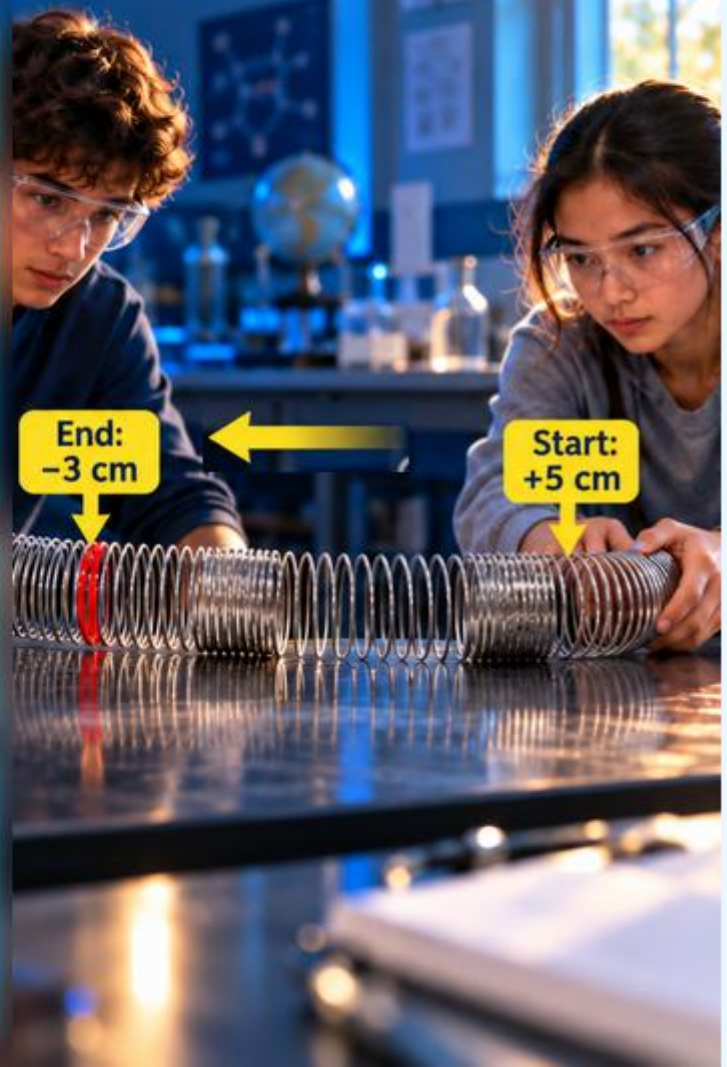
Students push and pull a slinky to make a longitudinal wave. The coils move back and forth in the same direction the wave travels. A painted coil starts 5 cm to the right of its rest spot. Then it moves to 3 cm left of its rest spot. Did it move 8 cm to the left?

CLUES

- Start: +5 cm
- End: −3 cm
- Change = end − start

COIL POSITION (cm)





CLAIM

Did the coil move 8 cm to the left?

☐ **YES** ☐ **NO**

EVIDENCE

Solve:

−3 − 5 = _____

Rewrite as adding: −3 + _____ = _____

REASONING

Finish the sentence:

−3 − 5 is the same as

−3 + _____ = _____.

So the coil moved _____ cm left.

4.06: Transverse Waves

Part 1: Make a Transverse Wave

Interactive Link 1
Waves made with springs (JavaLab)

Interactive Link 2
Wave Propagation (JavaLab)

★ Key Ideas

- A **wave** carries **energy** from place to place. The **medium** only moves in place.
- **Transverse** wave: the medium moves **up and down**, at right angles (perpendicular) to the wave.
- Transverse waves can be **mechanical** (they need a medium, like a rope or a guitar string)...
- ...or **electromagnetic** (EM), like light and radio waves. EM waves are always transverse and can cross empty space.

boxed = Unit 4 focus word

1 Open Link 1. Click **Up-Down**. Drag the red handle up and down. Watch one coil. Circle your answers. CIRCLE

USE LINK 1: WAVES MADE WITH SPRINGS

a) Each coil moves (up and down) (all the way to the end)

b) The wave moves (up and down) (to the right)

c) The coils move (parallel to) (at right angles to) the wave.

d) This is a (transverse) (longitudinal) wave.

2 Open Link 2. Check **Leaving traces**. Drag one dot and let go. Circle your answers. CIRCLE USE LINK 2: WAVE PROPAGATION

a) Each dot (moves across the screen) (wiggles in place)

b) After the wave passes, each dot (returns to its place) (stays at the end)

c) What travels across the screen? (energy) (the dots)

3 Drag a word into each blank. DRAG



A wave on a rope is a transverse _____ wave, because it needs a _____. Light is a transverse _____ wave. It can travel through empty _____.

(space) (medium) (longitudinal) (electromagnetic) (mechanical)

One extra word.

4 All of these are **transverse** waves. Which kind is each one? Circle **mechanical** (needs a medium) or **EM** (light family). CIRCLE

A spring shaken up and down	mechanical	EM
A plucked guitar string	mechanical	EM
Sunlight	mechanical	EM
A radio signal	mechanical	EM
A rope shaken up and down	mechanical	EM

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.06: Transverse Waves

Part 2: Transverse Waves in a Toy Spring

Interactive Link 1
Transverse Waves sim (Virtual Science Teachers)

Interactive Link 2
Properties of Waves diagram (Virtual Science Teachers)

★ Key Ideas

- A **transverse wave** looks like an up-and-down **S shape** with crests and troughs.
- **Frequency** changes how many waves pass each second. Higher frequency = shorter **wavelength**.
- **Amplitude** changes the height of the wave. Bigger amplitude = more **energy**.
- Neither one changes the wave speed in the same **medium** (100 cm/s here).

boxed = Unit 4 focus word

5 Open Link 1. Watch the toy spring. What does the shape look like? Circle your answers. CIRCLE USE LINK 1: TRANSVERSE WAVES SIM

a) The spring makes [an up-and-down S shape with crests and troughs] [bunched and spread-out coils]

b) The coils move [up and down] [back and forth] while the wave moves [to the right] [up and down]

c) This shape is a [transverse] [longitudinal] wave.

6 Move only the **Frequency** slider. Record the **Wavelength** at each frequency. FILL IN

Frequency	Wavelength
1 Hz	_____ cm
2 Hz	_____ cm
4 Hz	_____ cm
5 Hz	_____ cm

Frequency changes the [wavelength (how close the waves are)] [height of the waves]

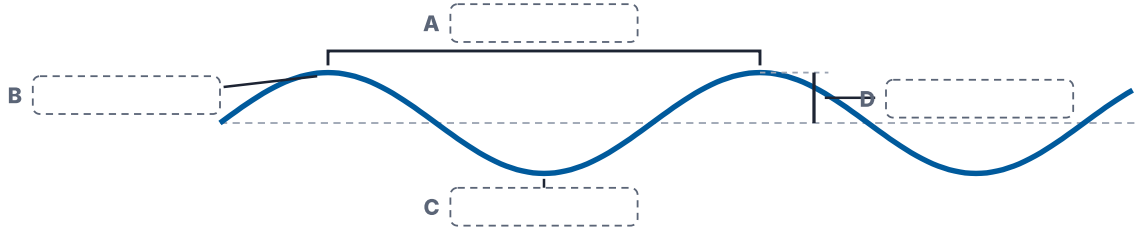
7 Now move only the **Amplitude** slider. Circle your answers. CIRCLE

a) Amplitude changes the [height of the waves] [wavelength]

b) A bigger amplitude carries [more] [less] energy.

c) The wave speed (100 cm/s) [changes] [stays the same]

8 Open Link 2 and finish the diagram. Then drag a word into each box on this wave. DRAG USE LINK 2: PROPERTIES OF WAVES DIAGRAM



[crest] [wavelength] [compression] [trough] [amplitude] *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #4 • THURSDAY

Is the diver over 50 m above the whale?

WAVE WORD: ***Mechanical wave***

A wave that needs a medium (such as water) to travel through.

MYSTERY

Sound is a mechanical wave, so it needs a medium to travel through. A whale sings 80 meters below the surface, and the song travels through the water. A diver hears it from 20 meters below the surface. Is the diver more than 50 meters above the whale?

CLUES

- Whale: −80 m
- Diver: −20 m
- Diver − whale





CLAIM

Is the diver more than 50 m above the whale?

☐ **YES** ☐ **NO**

EVIDENCE

Solve:

−20 − (−80) = _____

Rewrite as adding: −20 + _____ = _____

REASONING

Finish the sentence:

−20 − (−80) is the same as

−20 + _____ = _____.

The diver is _____ m above the whale.

4.07: Longitudinal Waves

Parts 1–2: Make a Longitudinal Wave

Interactive Link 1
Tuning Fork and Sound Wave (JavaLab)

Interactive Link 2
Longitudinal Waves sim (Virtual Science Teachers)

★ Key Ideas

- **Longitudinal** **wave**: the **medium** moves **back and forth**, in the same direction (parallel) as the wave.
- **Compression:** particles crowded together. **Rarefaction:** particles spread apart.
- **Sound** is a longitudinal, **mechanical** wave. It needs a **medium**.
- The particles only move in place. The **energy** moves along.

boxed = Unit 4 focus word

1 Open Link 1. Check **Basic motion of air molecules**. Watch the air as the tuning fork vibrates. Circle your answers. CIRCLE USE LINK 1: TUNING FORK

a) The fork pushes the air molecules [back and forth] [up and down]

b) The sound travels by bumping [air molecules] [empty space]

c) Sound is a [transverse] [longitudinal] wave.

2 Open Link 2 (the toy spring). Watch one coil. Circle your answers. CIRCLE USE LINK 2: LONGITUDINAL WAVES SIM

a) Each coil moves [up and down] [back and forth]

b) The coils move [parallel to] [at right angles to] the wave.

c) After the wave passes, each coil [returns to its place] [moves to the end]

3 Still in Link 2: set each **Frequency**. Record the **Wavelength** shown. FILL IN USE LINK 2

Frequency	Wavelength
2 Hz	_____ cm
4 Hz	_____ cm
5 Hz	_____ cm

Higher frequency = compressions [closer together] [farther apart]

4 Now move only the **Amplitude** slider. Circle your answers. CIRCLE USE LINK 2

a) Bigger amplitude: the compressions are [more crowded] [more spread out]

b) The wave carries [more] [less] energy.

c) The wavelength [changes] [stays the same]

d) One wavelength is from [compression to compression] [compression to rarefaction]

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.07: Longitudinal Waves

Part 3: Practice Wave Parts

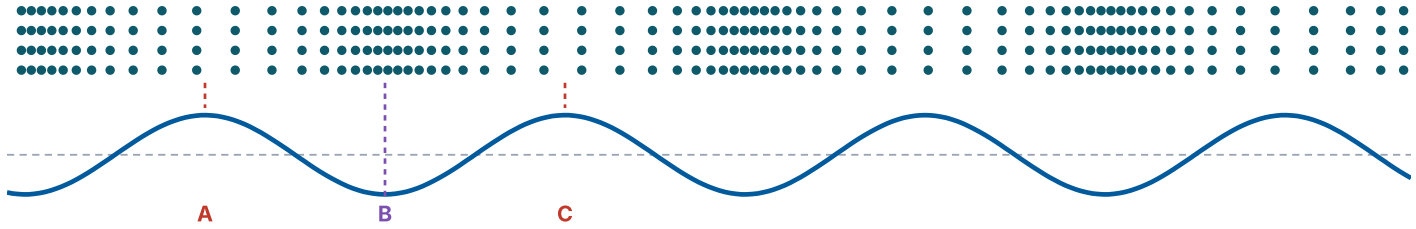
Interactive Link
Sound Wave Diagram (Virtual Science Teachers)

★ Key Ideas

- Scientists often draw sound as a graph. A **crest** shows a **compression**. A **trough** shows a **rarefaction**.
- A drawing with bunched-up lines is a **longitudinal** **wave**.
- **Transverse**: up and down, at right angles. Crests and troughs. Example: a shaken rope, light.
- **Longitudinal:** back and forth, parallel. Compressions and rarefactions. Example: sound.
- Longitudinal waves are always **mechanical**: they need a **medium**. EM waves (like light) are never longitudinal.

boxed = Unit 4 focus word

5 Open the Sound Wave Diagram. Drag each label onto it and press **Check**. Then circle your answers. CIRCLE USE THE SOUND WAVE DIAGRAM



a) At A and C, the particles are in a compression rarefaction. On the graph, that is a crest trough.

b) At B, the particles are in a compression rarefaction. On the graph, that is a crest trough.

c) One wavelength is from A to B A to C

d) A drawing with bunched-up lines is a transverse longitudinal wave.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.07 Quest Map

Review for the 4.07 Quest

Drag a word from the Answer Bank into each box. One extra word.

Need help? Look back at the Classkick slide for the lesson in each part's title. The questions here match the 4.07 Quest.

Answer Bank *One extra word.*

space | refraction | energy | wavelength | amplitude | short | distance | bend | decreases | medium | same

Part 1: Refraction Lesson 4.04

1. [ ] is the bending of light when it enters a different medium.
2. When light enters a denser medium, like air into glass, the speed of the wave [ ].
3. A straw looks broken in a glass of water. Light waves change speed and [ ] as they cross into a new medium.

Don't fall for it: Reflection is light *bouncing* off a surface. The straw does not really bend, and it is not resonance.

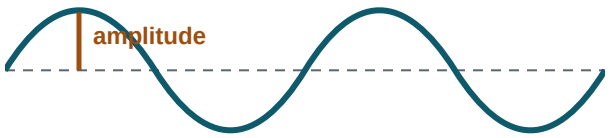
Part 2: Speed, Frequency, and Wavelength Lesson 4.03



4. Two waves travel through the same medium, water. The waves travel at the [ ] speed.
5. High frequency waves have [ ] wavelengths.

Don't fall for it: Frequency does not change speed. The medium changes speed.

Part 3: Amplitude and Energy Lesson 4.02



6. Amplitude is the maximum [ ] a wave moves from its rest position.
7. Two waves are the same except for amplitude. The wave with the larger amplitude has more [ ].

Don't fall for it: A bigger amplitude does NOT make a wave faster, and it does not change its frequency or wavelength.

Part 4: Light vs. Mechanical Waves Lessons 4.06, 4.07, 4.08

8. Light waves can travel through empty [ ]. Mechanical waves cannot.
9. Sound cannot travel in space because sound needs a [ ]. Light does not.
10. For a light wave, the [ ] decides its color.

Don't fall for it: Both light and sound have amplitude and wavelength, and both carry energy. The difference is that light does not need a medium.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

Friday is a school day! Spend at least 1 hour on science today.

Science Friday! Week 2 · Friday, Nov 20

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	CLASSKICK Flip the Sign + math Cases #1–#4	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.04 slides: Refraction + Refraction Lab	With 4.14 Test*	Yes / No
☐	GRADED **4.04 Refraction Lab** · 10 pts	**Closes Mon, Nov 23** ext: Mon, Nov 30	Yes / No
☐	CLASSKICK 4.05 Thunder Lab + graph, 4.06 Parts 1–2	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.07 Longitudinal Waves: Parts 1–3 + Quest Map	With 4.14 Test*	Yes / No
☐	GRADED **4.07 Quest** · 10 pts	**Closes Mon, Nov 30** ext: Thu, Dec 3	Yes / No

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished

Coming Up Next Week

Nov 23 – Nov 27 · Red tags show what **closes** that day.

MON 23	4.08 The Electromagnetic Spectrum **CLOSES: 4.04 REFRACTION LAB**
TUE 24	4.09 Review and Catch-Up
WED 25	*Thanksgiving Break · No School*
THU 26	*Thanksgiving Break · No School*
FRI 27	*Thanksgiving Break · No School*



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Science 7 · Week 3: Nov 23 – Nov 27

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · NOV 23

4.08: The Electromagnetic Spectrum

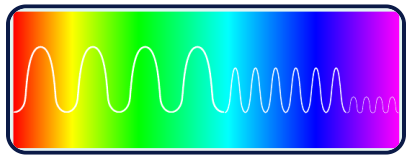
WARM-UP Integers: **The 4 Rules** + **Case #1** (storm alert).

READ Read lesson **4.08: The Electromagnetic Spectrum**.

CLASSKICK Complete the **4.08** Classkick slides: Part 1 and the **EM Spectrum Escape Room** (paste your certificate).

DUE! **4.04 Refraction Lab** is due today.

EXT DUE! Extended time: **3.12 Test** closes today.



TUESDAY · NOV 24

4.09: Review and Catch-Up

WARM-UP Integers: **Case #2** (the robot).

CLASSKICK Finish any 4.01–4.08 slides. Then play the wave games and the **Waves Choice Board** (integers + waves) on the 4.09 slides.

BONUS Extra time? Try the **Bonus Lab**: the Particle Wave.

REDO Lower than a B on 4.03 or 4.07? Review and redo it.



WEDNESDAY · NOV 25

No School

Thanksgiving Break
Enjoy your time off!

THURSDAY · NOV 26

No School

Thanksgiving Break
Happy Thanksgiving!

FRIDAY · NOV 27

No School

Thanksgiving Break
See you Monday!

INTEGER CREW: THE 4 RULES

Adding + Subtracting Integers

1 • SAME SIGNS? TEAM UP!

Add the counts. Keep the sign.



$$-6 + (-3) = -9$$

2 • DIFFERENT SIGNS? FACE OFF!

Subtract the counts. Keep the sign of the greater absolute value.

Greater absolute value = farther from zero.



$$-7 + 3 = -4$$

3 • SUBTRACT A POSITIVE? FLIP IT!

Keep the first. Add the opposite.



$$7 - 10 = 7 + (-10) = -3$$

4 • SUBTRACT A NEGATIVE? FLIP IT!

Keep the first. Add the opposite.



$$-9 - (-2) = -9 + 2 = -7$$

Subtract? KEEP the first. CHANGE to +. FLIP the second!

Robots represent signed numbers.

CASE #1 • MONDAY

Does the storm alert stay on?

WAVE WORD: *Infrared*

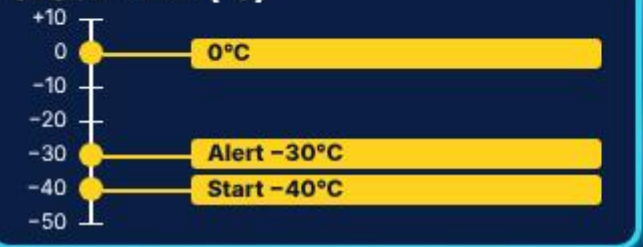
An EM wave with a longer wavelength than red light. We feel it as heat.

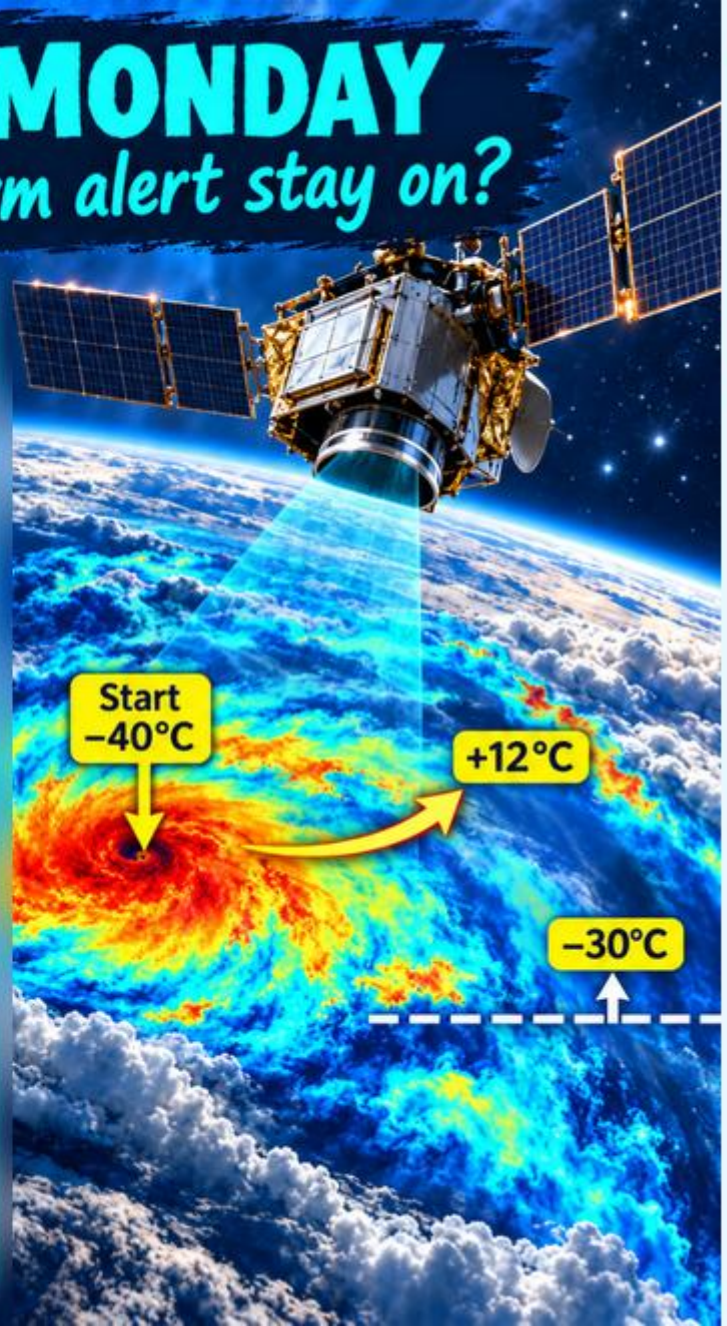
MYSTERY

Weather satellites use infrared waves, part of the electromagnetic spectrum, to measure how cold cloud tops are. A storm's cloud top is 40 degrees below zero (Celsius). As the storm weakens, the cloud top warms up 12 degrees. The storm alert stays on only while the cloud top is colder than −30°C. Does the alert stay on?

CLUES

- Start: −40°C
- Warms up: +12°C
- Alert: below −30°C





CLAIM

Does the storm alert stay on?

☐ YES ☐ NO

EVIDENCE

Solve:

−40 + 12 = _____

The cloud top is now _____ °C.

REASONING

Finish the sentence:

Different signs: I subtract 40 − 12 = _____ and keep the sign of _____.

−28°C is _____ than −30°C.

4.08: EM Spectrum

Part 1: The EM Spectrum

Interactive Link 1
EM Spectrum Drag and Drop (Virtual Science Teachers)

★ Key Ideas

- **Electromagnetic spectrum:** the whole range of EM waves. All travel through space at the **speed of light**.
- Order (long → short): **radio, microwave, infrared, visible, ultraviolet, X-ray, gamma**. **Visible light** is the only part we see.
- Radio waves can be as long as a **building**. Gamma rays are as small as an **atom**.
- Higher **frequency** = shorter **wavelength** = more **energy**.
- In a rainbow, **red** has the longest wavelength and **violet** the shortest.

boxed = Unit 4 focus word

1 Drag each wave into its place on the spectrum. ▶ Watch first: EM Spectrum video DRAG



◀ longest wavelength · lowest frequency · least energy

shortest wavelength · highest frequency · most energy ▶

Visible light: red = longest wavelength · violet = shortest

X-ray | visible light | sound | radio | gamma ray | infrared | ultraviolet | microwave *One extra word.*

2 Open Link 1. Drag each wave type onto the spectrum and press **Check**. Then drag a wave into each box below. DRAG USE LINK 1: EM SPECTRUM DRAG AND DROP

Clue from the spectrum	EM wave
Heat waves used in TV remotes and thermal cameras	
Short radio waves used for cooking and wireless signals	
Waves that cause sunburn and kill germs	
Waves for medical images and security scanners	
Highest-energy waves, used in medicine and given off in nuclear reactions	

X-rays | infrared | radio waves | gamma rays | ultraviolet | microwaves *One extra word.*

3 Use the size scale on Link 1. Circle your answers. CIRCLE

a) The longest waves are about the size of a **building** / **atom**

b) The shortest waves are about the size of an **apple** / **atom**

c) Gamma rays have the **lowest** / **highest** energy.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.08: EM Spectrum

Part 2: EM Spectrum Escape Room

Interactive Link
EM Spectrum Escape Room (Virtual Science Teachers)

4 **Escape the room, then paste your certificate.** SCREENSHOT

USE THE LINK ON THIS SLIDE

Step 1 Open the link. Play the **EM Spectrum escape room** from Virtual Science Teachers. Use your notes and Part 1 to solve each puzzle.

Step 2 Keep going until you escape. The **certificate is the LAST thing** in the escape room. It asks you to **type your name**.

Step 3 Take a screenshot of your certificate with the **Snipping Tool**: press **Windows + Shift + S** (Chromebook: **Ctrl + Shift + Show windows**) and drag a box around the certificate.

Step 4 Press **Ctrl + C** to copy it (if it did not copy already).

Step 5 Click the box on the right and press **Ctrl + V** to paste it onto this Classkick slide.



Paste your escape room certificate here

Your name must show on the certificate.

Snip it (Windows + Shift + S) · copy it (Ctrl + C) · paste it here (Ctrl + V)

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #2 • TUESDAY

Is the robot less than 25 m above the lake bottom?

WAVE WORD: *Sonar*

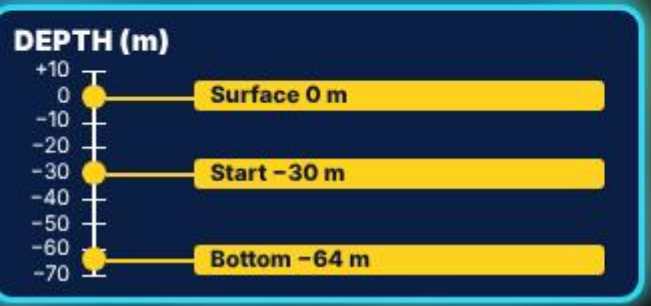
Using sound waves and their echoes to find things underwater.

MYSTERY

An underwater robot uses sonar, a sound wave, to map Lake Erie. It starts 30 meters below the surface, then dives 12 meters deeper. Its sonar finds the lake bottom 64 meters below the surface. Is the robot less than 25 m above the lake bottom?

CLUES

- Start: −30 m
- Dives: −12 m
- Bottom: −64 m





CLAIM

Is the robot less than 25 m above the lake bottom?

☐ **YES** ☐ **NO**

EVIDENCE

Solve:

1) −30 + (−12) = _____

2) −42 − (−64) = _____

Rewrite as adding: −42 + _____ = _____

REASONING

Finish the sentence:

Same signs add and keep the sign:

the robot is at _____ m. −42 + _____ = _____.

So the robot is _____ m above the bottom.

4.09: Review and Catch-Up

Catch up, then play the wave games

Interactive Link 1
Wave Combinator (Field Day)

Interactive Link 2
Wavemaker (Universe & More)

★ Key Ideas

- Waves carry **energy**, not matter.
- Describe a **wave** by its **speed**, **wavelength**, **amplitude**, and **frequency**.
- **Mechanical** waves need a **medium**. **EM** waves do not.
- **Transverse**: up and down. **Longitudinal:** back and forth.

boxed = Unit 4 focus word

1 Catch up! Check the box for each Classkick slide you have finished. CHECK

- ☐ 4.01 Describe a Wave (Parts 1–2)
- ☐ 4.02 Amplitude (Parts 1–2)
- ☐ 4.03 Frequency and Speed + CER + **4.03 Quest Map**
- ☐ 4.04 Refraction + Refraction Lab Map
- ☐ 4.05 Temperature Lab (Parts 1–2) + Graph (Part 3)
- ☐ 4.06 Transverse Waves (Parts 1–2)
- ☐ 4.07 Longitudinal Waves (Parts 1–3) + **4.07 Quest Map**
- ☐ 4.08 EM Spectrum + Escape Room certificate

Not done? Go back and finish it today. Lower than a B on the 4.03 or 4.07 Quest? Review your Quest Map, then redo it.

2 The big idea of Unit 4 (review 4.01, 4.06, 4.07). Circle your answers. CIRCLE

> Vibrations cause waves that transfer energy from one place to another. The medium moves for a moment as the energy passes, then returns to where it started.

a) Does **energy** move from place to place in a wave? YES NO

b) Does the **matter** (the medium) travel along with the wave? YES NO

c) A ribbon is tied to a rope. A wave passes. The ribbon moves up and down / rides to the end

3 Link 1: Wave Combinator (review 4.02, 4.03). Change the waves to match the target. Then drag a word into each blank. DRAG USE THE LINK ON THIS SLIDE

To make the wave taller, I changed the _____. To fit more waves in the same space, I made the wavelength _____.

When two waves add together, a crest plus a crest makes a _____ wave.

shorter | amplitude | longer | bigger

One extra word.

Highest level I reached: _____

4 Link 2: Wavemaker (review 4.02, 4.03, 4.06). Shake the string to match the target shape. Circle what worked. CIRCLE

More waves on the string: I shook faster / slower

That means a higher frequency / amplitude

A taller wave: I shook bigger / smaller

All done? Try **The Case of the Missing Cell Phone** (review 4.02, 4.08) or the **Bonus Lab: The Particle Wave** (review 4.03) on the next slides.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.09: Review and Catch-Up

Part 2: Waves Choice Board (Integers + Waves)

Interactive Link
Waves Choice Board

★ Key Ideas

- **Adding a negative** = subtracting: 3 + (−5) = 3 − 5 = −2.
- **Subtracting a negative** = adding: 4 − (−4) = 4 + 4 = 8.
- Rest position = **0**. Above is **+** (crest side). Below is **−** (trough side).
- Change = **end − start**. A negative change means it went down.

boxed = Unit 4 focus word

5 Open the **Waves Choice Board**. Finish 3 squares in a row (across, down, or diagonal). Check the 3 you finished. CHECK

USE THE CHOICE BOARD

☐ 1. Wave Anatomy	☐ 2. Wavelength	☐ 3. Frequency
☐ 4. Mediums	☐ 5. CER Mystery	☐ 6. Wave Motion
☐ 7. Wave Types	☐ 8. Mechanical Waves	☐ 9. EM Spectrum

Each square has 1 integer problem + 1 science question. It checks your answers for you.

6 Integers and waves. Fill in each answer. FILL IN

a) A crest is at **+5 cm**. A trough is at **−5 cm**. Crest − trough = _____

b) A rope moves from **+3 cm** down to **−2 cm**. Change = −2 − (+3) = _____

c) Air warms from **−6 °C** to **+9 °C**. Change = 9 − (−6) = _____

d) Does the warmer air make sound go faster slower?

Finish the board: paste your certificate

📸

Snip your Congratulations screen and paste it here

Windows + Shift + S (Chromebook: Ctrl + Shift + Show windows), then Ctrl + V here.

7 Done early? Review the Unit 4 vocabulary.

🎮 **Unit 4 Vocab: Blooket**

🎮 **Unit 4 Vocab: Gimkit**

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

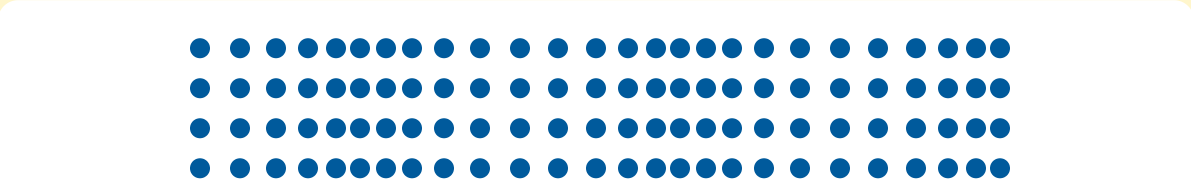
Bonus: Extra Practice

Optional lab for 4.09 Review Day

Review Day Bonus Lab

This lab is **optional**. Do it on **4.09 Review Day** after you finish your catch-up work and the two wave games. It is great practice for the Unit 4 test.

Bonus Lab: The Particle Wave



Watch the particles in a wave. Measure the wavelength and see how frequency changes it.

Next slide.

Remember: Bonus slides are not required. Skip them if you are still catching up.

Bonus Lab

The Particle Wave: Does Wavelength Change Wave Speed?

Interactive Link
The Particle Wave (Physics Classroom)

★ Key Ideas

- Waves in the **same** **medium** travel at the **same speed**.
- **Shorter** **wavelength** = more waves pass each second = **higher** **frequency**.
- **Amplitude** is the height of the **wave**. Bigger amplitude = more **energy**.
- **speed = wavelength × frequency**

boxed = Unit 4 focus word

Lab Steps

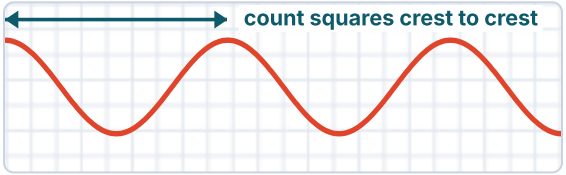
Set up

- Open the link. Click **Start**.
- Click **Real Time**.
- Leave **Speed** at 4.00 m/s and **Amplitude** at 0.80 m.
- Only change the **Wavelength**.

Test each wavelength (1.00 m, 2.00 m, 3.00 m)

- Type the wavelength.
- Press play ▶, then pause.
- Find a crest (the top of the wave).
- Count the squares to the next crest.
- Record it in the table.
- Press play. Watch how fast Particle 12's **Cycle Counter** counts.

Then change only the **Amplitude** and watch the wave.



1 Record your data. FILL IN USE THE LINK ON THIS SLIDE

Wavelength (m)	Squares crest to crest	Speed (m/s)
1.00		
2.00		
3.00		

Tip: Use Slow Mo if the counter moves too fast to see.

2 Did changing the wavelength change the wave speed? YES / NO

YES NO

3 Which wavelength made Particle 12 count cycles the fastest (highest frequency)? CIRCLE

1.00 m 2.00 m 3.00 m

4 Change only the Amplitude. What changes? CIRCLE

height of the waves distance between crests wave speed

5 Use your data. When the speed stays the same, how are wavelength and frequency related? Drag a word into each blank. DRAG

The speed stayed the _____. When the wavelength was _____, Particle 12 went up and down _____ times each second. So a shorter wavelength has a _____ frequency.

shorter higher same more longer *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

No school Wed–Fri! Use this checklist on Tuesday.

Science Friday! Week 3 · Before break · Tue, Nov 24

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	CLASSKICK The 4 Rules + math Cases #1–#2	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.08 slides + Escape Room certificate	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.09 Review + Waves Choice Board	With 4.14 Test*	Yes / No
☐	CLASSKICK Bonus Lab: The Particle Wave (optional)	With 4.14 Test*	Yes / No
☐	GRADED **4.04 Refraction Lab** · 10 pts	Closes Mon, Nov 23 ext: Mon, Nov 30	Yes / No
☐	GRADED **4.07 Quest** · 10 pts	Closes Mon, Nov 30 ext: Thu, Dec 3	Yes / No

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished

Coming Up Next Week

Nov 30 – Dec 4 · Red tags show what **closes** that day.

MON 30	4.10 Wave Interference CLOSES: 4.07 QUEST CLOSES: EXT: 4.03 QUEST, 4.04 LAB
TUE 1	4.11 Sound Waves Lab
WED 2	4.12 Explore Sound in a Medium
THU 3	4.13 Unit 4 Review CLOSES: EXT: 4.07 QUEST
FRI 4	Science Friday!



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Science 7 · Week 4: Nov 30 – Dec 4

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · NOV 30
4.10: Wave Interference

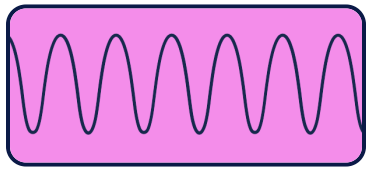
WARM-UP Integers: **Power Up!** + **Case #1** (ripples).

READ Read lesson **4.10: Wave Interference**.

CLASSKICK Complete the **4.10** Classkick slides: Part 1 and the **Interference Missions** game.

DUE! **4.07 Quest** is due today.

EXT DUE! Extended time: **4.03 Quest** and **4.04 Refraction Lab** close today.



TUESDAY · DEC 1
4.11: Sound Waves Lab

WARM-UP Integers: **Case #2** (decibels).

READ Read lesson **4.11: Sound Waves Lab**.

CLASSKICK Complete the **4.11** Classkick slides: Parts 1–2 (PhET Sound Waves).



WEDNESDAY · DEC 2
4.12: Explore Sound in a Medium

WARM-UP Integers: **Case #3** (lightning).

READ Read lesson **4.12: Explore Sound in a Medium**.

CLASSKICK Complete the **4.12** Classkick slides: Part 1 and the **Wave Speed Lab**.



THURSDAY · DEC 3
4.13: Unit 4 Review

WARM-UP Integers: **Case #4** (the seiche).

CLASSKICK **4.13 Review** slide, the **Vocab Review** (2 slides), and the **4.14 Mission Map** (we do it together in class).

REVIEW Play the **Unit 4 Wordwall** review game on the 4.13 page.

SURVEY Take the **Unit 4 vocabulary post test**. Closes Thu, Dec 10.

FRIDAY · DEC 4
Science Friday

FINISH Finish any lessons, Classkick slides, or assignments you have not completed this week.

REDO Check your grades. Lower than a B? Review the Classkick notes and redo it.

CLASSKICK Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.

STUDY Review your Mission Map and Unit 4 slides for Monday's test.

INTEGER CREW: POWER UP!

Multiplying Integers

The crew's power gate is locked! Match the signs to choose the charge.

SAME signs? POSITIVE power! | DIFFERENT signs? NEGATIVE power!

SAME SIGNS → POSITIVE

Multiply the counts. Make it positive.



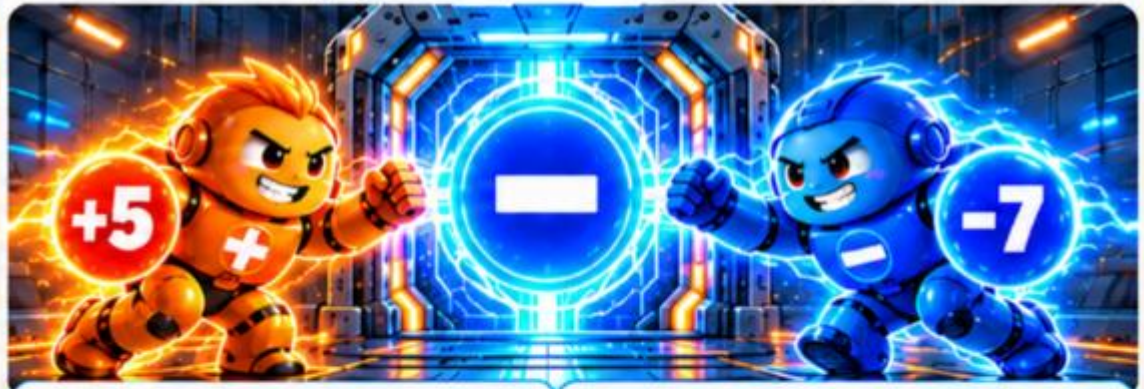
$(-6) \times (-2) = 12$



$6 \times 2 = 12$

DIFFERENT SIGNS → NEGATIVE

Multiply the counts. Make it negative.



$5 \times (-7) = -35$

$(-5) \times 7 = -35$

REPEATED ADDITION: $3 \times (-4) = (-4) + (-4) + (-4) = -12$

Any number × 0 = 0. Robots represent signed numbers.

CASE #1 • MONDAY

Is the trough deeper than −5 cm?

WAVE WORD: *Interference*
When two or more waves meet and combine.

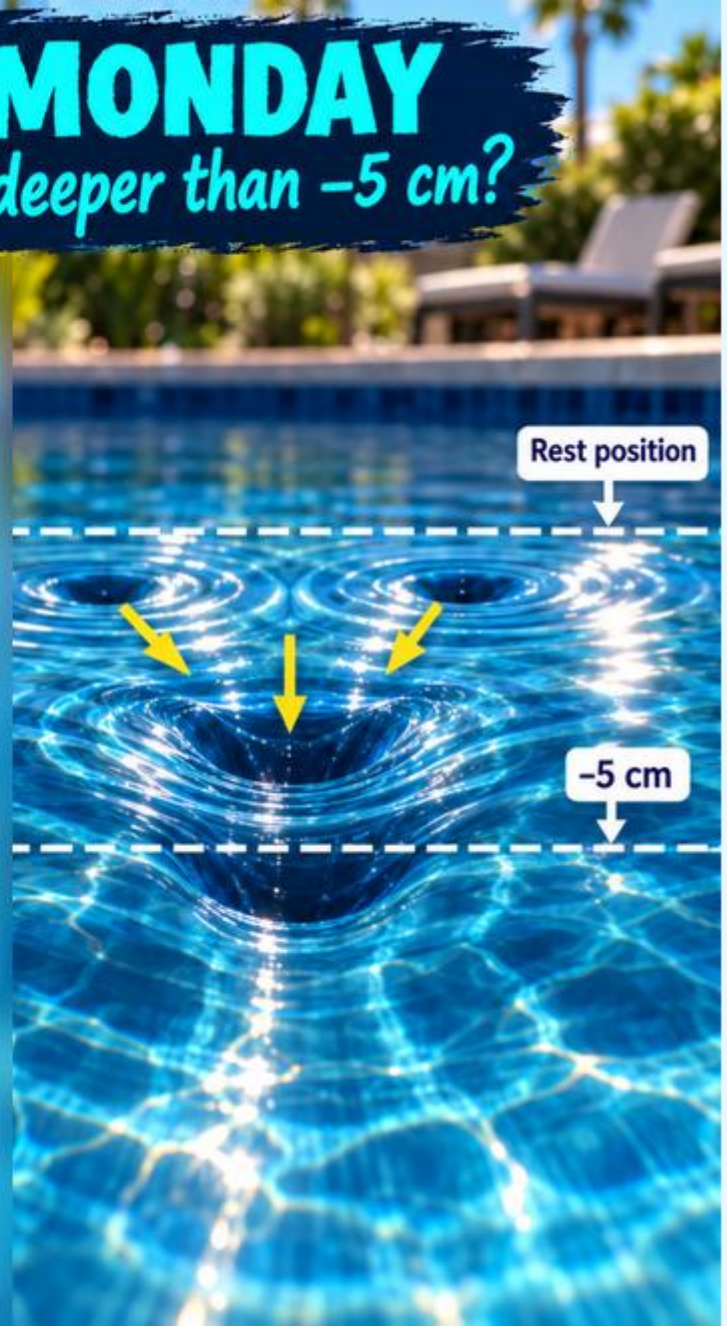
MYSTERY

Three ripples move across a pool. Each ripple has a trough that is 2 cm below the rest position. When all three troughs meet at the same spot, constructive interference adds them together into one deeper trough. Is the new trough deeper than −5 cm?

CLUES

- Ripples: 3
- Each trough: −2 cm
- Check: −5 cm





CLAIM

Is the new trough deeper than −5 cm?

☐ YES ☐ NO

EVIDENCE

Solve:

$3 \times (-2) =$ _____

Repeated addition: (−2) + (−2) + (−2) = _____

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. The new trough is at _____ cm. −6 cm is _____ than −5 cm.

4.10: Wave Interference

What Happens When Waves Meet?

Interactive Link
Wave Interference Video

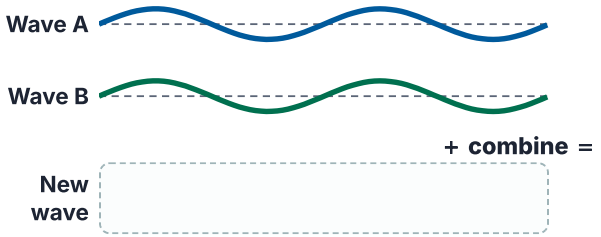
★ Key Ideas

- **Wave interference:** two or more waves overlap and combine to make a new wave pattern.
- **Constructive:** crest meets crest (in phase). The waves add up to a **bigger** wave.
- **Destructive:** crest meets trough (out of phase). The waves make a **smaller** wave or cancel out.
- After they meet, the waves keep going on their own paths, unchanged.

boxed = Unit 4 focus word

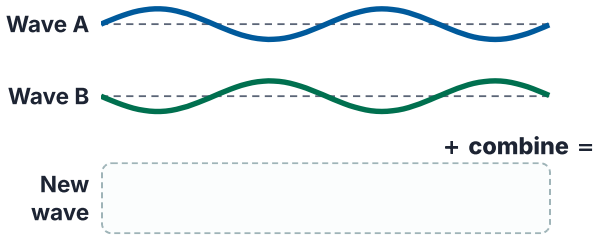
1 Watch the video. Then drag a word under each picture. DRAG USE THE LINK ON THIS SLIDE

Crest meets crest (in phase)



_____

Crest meets trough (out of phase)



_____

Word bank: destructive | refraction | constructive *One extra word.*

2 Two waves meet. Drag the amplitude of the new wave. DRAG

Wave A	Wave B	New amp.
3 in. crest	3 in. crest	
2 units crest	3 units crest	
4 units crest	4 units crest	
2 in. up (crest)	2 in. down (trough)	

5 | 1 | 8 | 0 | 6

3 Circle the type of interference. CIRCLE

Noise-canceling headphones quiet the sound around you.	constructive	destructive
Two ocean waves line up and make a bigger wave for surfing.	constructive	destructive
Two friends push a swing at the same time, and it goes higher.	constructive	destructive
Your radio has poor reception in one spot.	constructive	destructive

4 What is the difference between constructive and destructive interference? Drag a word into each blank. DRAG

Constructive: a crest meets a _____, so the amplitudes add and the new wave is _____.

Destructive: a crest meets a _____, so the new wave is _____ or flat.

smaller | crest | faster | trough | bigger *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.10: Wave Interference

Part 2: Interference Missions

Interactive Link
Interference Missions game

★ Key Ideas

- **Constructive:** crest meets crest. Amplitudes **add**.
- **Destructive:** crest meets trough. Amplitudes **subtract**.
- Same size + crest to trough = **0** (silence).

boxed = Unit 4 focus word

5 **Open the Interference Missions game. Check off each mission as you finish it.** CHECK **USE THE LINK ON THIS SLIDE**

- ☐ 1. Airplane Hush
- ☐ 2. Surf's Up
- ☐ 3. Harbor Engineer
- ☐ 4. Lawn Mower Level
- ☐ 5. Road Trip Radio
- ☐ ★. Lightning Round

6 **What did you see in the game? Circle your answers.** CIRCLE

a) Headphones went silent when crest met **crest** **trough**

b) Swells of 2 m and 3 m lined up crest to crest made a **1 m** **5 m** wave

c) For the louder mower, your wave also had to match its **amplitude** **color**

d) A radio dead spot is **constructive** **destructive** interference

7 **Finish Interference Missions. Type your completion code in the box.** TYPE

Paste a screenshot of your Certificate of Completion here

In the game: Copy certificate, then Ctrl + V here. Or take a screenshot (Windows + Shift + S, Chromebook: Ctrl + Shift + Show windows).

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #2 • TUESDAY

Was the sound quieter 5 seconds ago?

WAVE WORD: *Decibel*

The unit for measuring how loud a sound is (dB).

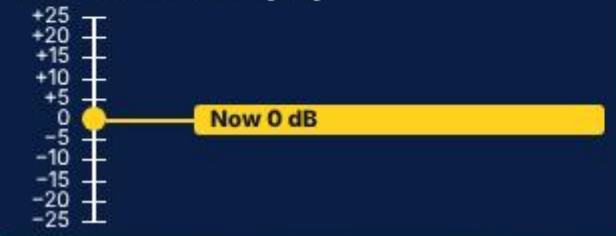
MYSTERY

In the sound lab, a student walks away from a speaker. The sound level changes by −4 decibels every second as she moves away. Her partner says that 5 seconds ago (−5 s), the sound was 20 decibels quieter than it is now. Is the partner right?

CLUES

- −4 dB per second
- Time: −5 s (the past)
- Partner says: quieter

SOUND VS. NOW (dB)





CLAIM

Is the partner right?

☐ **YES** ☐ **NO**

EVIDENCE

Solve:

(−4) × (−5) = _____

5 seconds ago, the sound was _____ dB _____.

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. 5 seconds ago the sound was _____ dB _____ than now.

4.11: Sound Waves Lab

Part 1: Music Lab Sound Waves

Interactive Link
Music Lab: Sound Waves

★ Key Ideas

- **Sound** is made by **vibrations**. It travels as a longitudinal **wave** through a **medium**.
- Each particle only moves **back and forth**, then returns to its place. The **energy** moves on.
- **Higher pitch** = higher **frequency** = shorter **wavelength**.
- **Louder** = bigger **amplitude** = more energy.
- Sound travels fastest in **solids** (like steel).

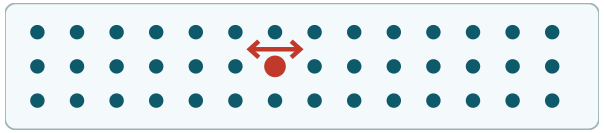
boxed = Unit 4 focus word

1 Open **Music Lab: Sound Waves** (works best in Chrome). Play a **low** note, then a **high** note. Circle what you see. CIRCLE USE THE LINK ON THIS SLIDE

	Low note	High note
Pitch	low high	low high
Particles vibrate	slower faster	slower faster
Compressions are	closer farther apart	closer farther apart
Wavelength	shorter longer	shorter longer

A higher pitch means a higher frequency bigger amplitude.

2 Watch the red particle in Music Lab. Drag a word into each blank. DRAG



The red particle moves _____ and then returns to _____. The sound _____ moves across the screen.

energy back and forth its place the end *One extra word.*

3 Watch the **blue dots** in Music Lab. Circle your answers. CIRCLE

a) The dots move
up and down back and forth

b) They move in the
same direction as at right angles to the sound wave.

c) So sound is a
transverse longitudinal wave.

4 A sound gets higher. Drag a word into each blank. DRAG

A higher sound has a _____ frequency, so the particles vibrate back and forth more _____. The compressions are closer together, so the wavelength is _____.

shorter often higher longer *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.11: Sound Waves Lab

Part 2: Sound Waves Lab (PhET)

Interactive Link
PhET Sound Waves

★ Key Ideas

- Higher **frequency** = rings closer together = shorter **wavelength** = higher pitch.
- Bigger **amplitude** = more **energy** = louder. It does not change the wavelength.
- **Wavelength:** measure from one crest (light ring) to the next.
- Two waves can add up (**constructive**) or cancel (**destructive**).
- Sound needs a **medium**. No air = no sound.

boxed = Unit 4 focus word

5 **Intro screen:** Check **Listener Audio**. Slide the **frequency** up. CIRCLE USE PHET SOUND WAVES

a) The rings get **closer together** / **farther apart**

b) The wavelength gets **shorter** / **longer**

c) The pitch sounds **higher** / **lower**

6 **Intro screen:** Now slide the **amplitude** up. CIRCLE

a) The rings look **darker and brighter** / **closer together**

b) The sound gets **louder** / **higher**

c) The wave carries **more** / **less** energy.

d) The wavelength **changes** / **stays the same**

7 **Measure screen:** Pause. Drag the ruler from one light ring to the next. FILL IN

Frequency	Wavelength (m)
200 Hz	
500 Hz	
1000 Hz	

Higher frequency = **shorter** / **longer** wavelength

8 **Two Sources screen:** Drag the girl up and down. Listen. CIRCLE

a) In some spots the sound is loud. Crest meets crest. That is **constructive** / **destructive** interference.

b) In quiet spots, crest meets **crest** / **trough**. That is **constructive** / **destructive** interference.

9 **Air Pressure screen:** Slide **Air Density in Box** all the way down. CIRCLE

a) The waves leaving the box **fade away** / **get bigger**

b) With no air (a vacuum), sound **can** / **cannot** travel.

c) Sound needs a **medium** / **light source**

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #3 • WEDNESDAY

Is the lightning more than 1 km away?

WAVE WORD: *Speed of sound*

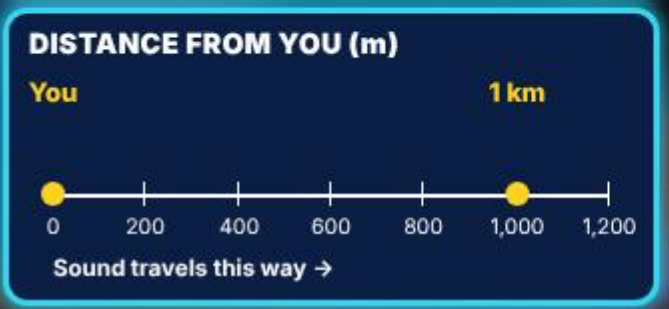
How fast sound travels: about 340 m/s in air.

MYSTERY

You see a lightning flash, then hear thunder 3 seconds later. Light reaches you almost instantly, but sound travels through the air at about 340 meters per second. Distance = speed × time. Is the lightning more than 1 km away?

CLUES

- Speed: 340 m/s
- Time: 3 s
- 1 km = 1,000 m





CLAIM

Is the lightning more than 1 km away?

☐ YES ☐ NO

EVIDENCE

Solve:

340 × 3 = _____

The lightning is _____ m away.

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. 1,020 m is _____ than 1,000 m (1 km).

4.12: Explore Sound in a Medium

How Does Sound Move Through a Medium?

Interactive Link
Visualizing Sound in a Medium (OpenSciEd)

★ Key Ideas

- **Sound** is a mechanical **wave**. It needs a **medium** (solid, liquid, or gas).
- Sound is a **longitudinal** wave: particles vibrate back and forth in the same direction the wave moves.
- **Compression:** particles pushed close. **Rarefaction:** particles spread apart.
- **Loudness** = **amplitude** (more **energy**). **Pitch** = **frequency**.
- Sound is usually fastest in **solids**, then liquids, slowest in gases.

boxed = Unit 4 focus word

1 Open the simulation and press play. Move one slider at a time. Circle what you see. CIRCLE USE THE LINK ON THIS SLIDE

I moved...	Motion of a Sound Source graph	Particle diagram
LOUDNESS: soft → loud	The graph gets **taller** **shorter**	The particles get pushed **closer together** **less together**
PITCH: low → high	Waves in 1 second: **more** **fewer**	Bunches of particles are **closer together** **farther apart**

Loudness is controlled by **amplitude** **frequency**

Pitch is controlled by **amplitude** **frequency**

The particles **travel to the end** **vibrate back and forth**

Sound is a **transverse** **longitudinal** wave

2 These graphs show two sounds. Circle the answers. CIRCLE

Sound A



Sound B



Louder sound: **A** **B**

Higher pitch: **A** **B**

More energy: **A** **B**

3 Why is sound faster in a solid than in air? Why is there no sound in space? Drag a word into each blank. DRAG

In a solid, the particles are packed _____, so each one bumps the next one quickly. Space has _____, so there is no _____ to carry the sound.

no particles **far apart** **medium** **close together** *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.12: Explore Sound in a Medium

Part 2: Wave Speed Lab

Interactive Link
Wave Speed Lab

★ Key Ideas

- **Wave speed** depends on the **medium** the wave travels through.
- **Frequency**, **wavelength**, and **amplitude** do NOT change wave speed.
- Same medium = same speed, high pitch or low.
- Sound is faster in **water** and **steel** than in air.

boxed = Unit 4 focus word

4 Open the Wave Speed Lab. Check off each mission as you finish it. CHECK USE THE LINK ON THIS SLIDE

- ☐ 1. Trampoline Frequency
- ☐ 2. Who Will Win? (predict)
- ☐ 3. The Big Race
- ☐ 4. Hops Make Waves
- ☐ 5. Rope Race (4 rounds)
- ☐ 6. Same Note, New Medium
- ☐ 7. Tin Can Phones
- ☐ ★. Fact or Myth?

5 What did you see? Circle your answers. CIRCLE

a) The clap went from air into water. In the water it **sped up** **slowed down** **stayed the same**

b) The same 440 Hz note was fastest in **air** **water** **steel**

c) Squeak's and Rumble's "HELLO!" arrived **at the same time** **high voice first**

d) Kiki hops more often than Bo. Kiki is **faster** **the same speed**

6 What changes the speed of a wave? What does NOT? Use an example from the lab. Drag a word into each blank. DRAG

Only the _____ changes wave speed. The clap sped up when it went from air into _____. Changing the _____ does not change speed: Squeak and Rumble arrived _____.

medium **together** **frequency** **water** **amplitude**

One extra word.

7 Finish the Wave Speed Lab. Type your completion code in the box. TYPE

📸

Paste a screenshot of your Certificate of Completion here

In the game: Copy certificate, then Ctrl + V here. Or take a screenshot (Windows + Shift + S, Chromebook: Ctrl + Shift + Show windows).

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

CASE #4 • THURSDAY

Did Toledo's water drop more than 10 feet?

WAVE WORD: *Seiche*

A giant wave that sloshes back and forth across a lake.

MYSTERY

Strong winds can push Lake Erie's water from one end to the other, making a giant sloshing wave called a seiche. During a storm, the water level at Toledo changes by −2 feet every hour for 4 hours. Did the water drop more than 10 feet?

CLUES

- Each hour: −2 ft
- Hours: 4
- Check: 10 ft





CLAIM

Did the water drop more than 10 feet?

☐ YES ☐ NO

EVIDENCE

Solve:

(−2) × 4 = _____

The water level changed by _____ ft.

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. The water dropped _____ ft, which is _____ than 10 ft.

4.13: Unit 4 Review

Review: All 10 Focus Words

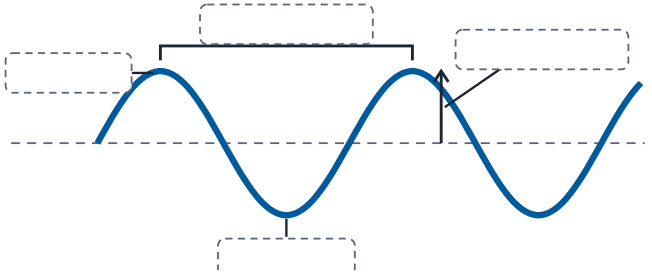
Interactive Link
Review Waves and EMS (Wordwall)

★ Key Ideas

- **Wave**: carries **energy**, not matter.
- **Transverse wave**: the **medium** moves up and down, at a right angle to the wave. **Longitudinal:** back and forth (sound).
- Mechanical waves need a **medium**. An **electromagnetic wave** (light) can travel through empty space.
- Bigger **amplitude** = more energy. Higher **frequency** = shorter **wavelength**.
- **Refraction**: a wave bends when its speed changes in a new medium.
- **Density** flip: light is slower in denser stuff; sound is faster in solids.

boxed = Unit 4 focus word

1 Drag the labels onto the wave. DRAG



crest | trough | wavelength | amplitude | frequency

One extra word.

2 Drag the focus word that matches each meaning. DRAG

_____ A disturbance that carries energy from place to place.

_____ The matter a wave travels through (solid, liquid, gas).

_____ How many waves pass a point each second.

_____ The ability to do work or cause change.

_____ How much mass is packed into a space.

density | energy | refraction | wave | frequency | medium

One extra word.

3 Circle both words for each wave. CIRCLE

Wave	Motion	Type
Rope wave	transverse / longitudinal	mechanical / EM
Sound	transverse / longitudinal	mechanical / EM
Sunlight	transverse / longitudinal	mechanical / EM
Microwaves	transverse / longitudinal	mechanical / EM

4 Density and refraction. Circle the answers. CIRCLE

a) Light goes from air into water. It **speeds up** / **slows down** and **bends** / **goes straight**

b) Bending of a wave in a new medium is **refraction** / **interference**

c) Sound travels fastest in **air** / **water** / **steel**

d) Light travels fastest in **air** / **water** / **glass**

5 Play the Wordwall review. Then explain: why can we see the Sun but not hear it? Drag a word into each blank. DRAG

USE THE LINK ON THIS SLIDE

Light is an _____ wave, so it can travel through empty space. Sound is a _____ wave and needs a _____.

longitudinal | electromagnetic | mechanical | medium

One extra word.

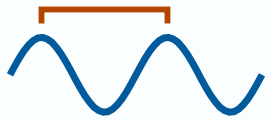
Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

Unit 4 Vocab Review

Part 1: Picture Match

Interactive Link
Wave Vocab Flashcards (SERP)

1 Use the picture and the meaning. Drag the focus word into each box. DRAG



The distance from one crest to the next crest.

_____



The bending of a wave when it enters a new medium.

_____



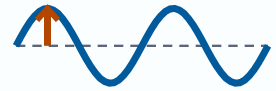
A disturbance that carries energy from one place to another.

_____



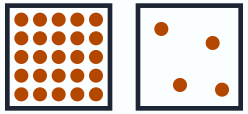
The matter a wave travels through.

_____



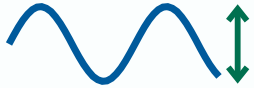
How far the medium moves from rest. Bigger = more energy.

_____



How much mass is packed into a space.

_____



The medium moves at a right angle to the wave direction.

_____



The ability to do work or cause change. Waves carry it.

_____



A wave that can travel through empty space, like light.

_____



How many waves pass a point each second.

_____

Word bank: density | medium | wave | refraction | frequency | energy | amplitude | transverse wave | crest | wavelength | electromagnetic wave

One extra word.

SURVEY **Don't forget!** Take the **Unit 4 Post Vocab** survey in Canvas today. Rate each word again to show how much you grew. It closes **Thu, Dec 10**.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

Unit 4 Vocab Review

Part 2: Use the Words

Interactive Link
Wave Diagram Labeling (Wordwall)

2 **Odd one out! Circle the word that does NOT belong.** CIRCLE

light | microwaves | X-rays | sound

crest | trough | density | wavelength

solid | amplitude | liquid | gas

ocean wave | rope wave | sound | guitar string

Be ready to say **why** it does not belong.

3 **At the concert. Drag a focus word into each blank.** DRAG

A guitar string vibrates and makes a sound _____. It carries _____ through the air, which is the _____. Plucking harder makes a bigger _____, so the sound is louder. A tighter string vibrates faster, so the _____ goes up and the _____ gets shorter. The stage lights give off light, an _____. When the light passes into a glass of water, it bends. This is _____.

energy | refraction | wave | density | wavelength | medium | amplitude | electromagnetic wave | frequency *One extra word.*

4 **Pick ONE focus word. Explain it so a 5th grader would understand. Give an example.** TYPE

5 **Now take the Unit 4 Post Vocab survey. Rate each word again.** CHECK

☐ I finished Part 1 and Part 2.

☐ I took the Post Vocab survey.

Circle the words you can now explain to everyone:

wavelength | refraction | wave | medium | amplitude | density | transverse wave | energy | electromagnetic wave | frequency

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.14 Unit 4 Mission Map

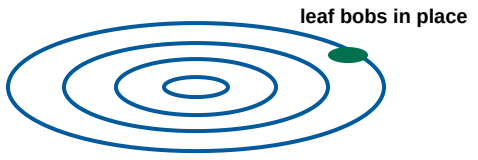
Notes for the Unit 4 Mission (test)

Each part has its own Word Bank.
Drag a word from that bank into each box.

Need help? Each part matches a group of Mission questions (the numbers are in the part title). Look back at your Classkick slides for 4.01–4.13 and the Unit 4 Review.

SURVEY **Don't forget!** Take the **Unit 4 Post Vocab** survey in Canvas today. Rate each word again to show how much you grew. It closes **Thu, Dec 10**.

Part 1: Waves Carry Energy Questions 2, 3, 11



Word Bank: new position | in place | energy | same position | increases

One extra word.

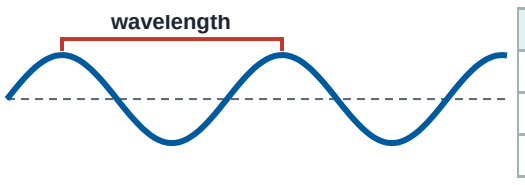
2. The water level rises and falls, and a leaf bobs in place. Across the pond, _____ is transferred. The water molecules only move up and down _____.

11. After a wave passes through a spring, each coil returns to about the _____.

3. A loud bass note has a high amplitude. This _____ the energy of the wave.

Don't fall for it: Waves move **energy**, not matter. Amplitude does not change the speed or the type of wave.

Part 2: Parts of a Wave Questions 1, 4



Wave	Amplitude	Frequency	Wavelength
A	2 cm	5 Hz	10 cm
B	**4 cm**	5 Hz	10 cm
C	2 cm	10 Hz	5 cm

Word Bank: amplitude | trough | crest | energy

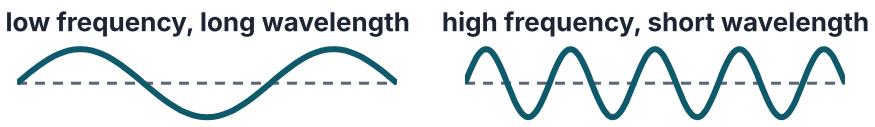
One extra word.

1. Wavelength is the distance from _____ to crest, or trough to trough.

4. Wave B has a greater _____, so it carries more _____.

Don't fall for it: Crest to **trough** is not a wavelength. Wave C has a higher **frequency**, not a bigger amplitude.

Part 3: Frequency and Wavelength Questions 5, 6, 18



Word Bank: increases | quickly | decreases | slowly

One extra word.

5. In a ripple tank with the same depth everywhere, when the frequency goes up, the wavelength _____.

6. As the wavelength decreases, the frequency _____.

18. For a high-pitched (high-frequency) sound, air particles vibrate back and forth more _____.

Don't fall for it: Frequency and wavelength go in **opposite** directions. One goes up, the other goes down.

Part 4: Kinds of Waves Questions 9, 10, 12



Word Bank: longitudinal | sunlight | sound | earthquake | up and down

One extra word.

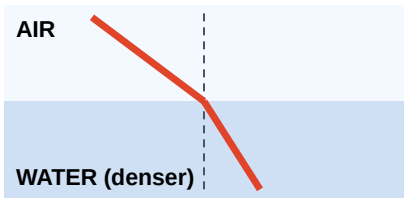
9. In a transverse wave, particles move _____.

10. A wave drawn with bunched-up vertical lines is _____.

12. Two mechanical waves that need a medium: _____ from a guitar and an _____ wave in rock.

Don't fall for it: Sunlight and radio waves are **electromagnetic**. They do not need a medium.

Part 5: Refraction and Density Questions 7, 8, 17



Sound speed:
steel 5,960 m/s
water 1,480 m/s
air 343 m/s

Word Bank: close together | increases | decreases | speed

One extra word.

7. Light bends in water because its _____ changes in the denser medium.

8. As density increases, the speed of light _____.

17. Sound is fastest in steel because its particles are packed _____.

Don't fall for it: The **frequency** (color) of light does not change in water. Light is *slower* in denser stuff, but sound is *faster* in solids.

Part 6: EM Waves and Space Questions 13, 14, 15, 16



Word Bank: water | medium | frequency | decreases | amplitude | increases | wavelength | vacuum

One extra word.

13. From radio to gamma, wavelength _____ and frequency _____.

14. Gamma rays carry more energy because they have a higher _____ and a shorter _____.

15. A probe cannot send sound through space. Mechanical waves need a _____.

16. Radio waves travel through the _____ of space. Sonar needs _____.

Don't fall for it: Gamma rays are not faster. All EM waves travel at the same speed in space.

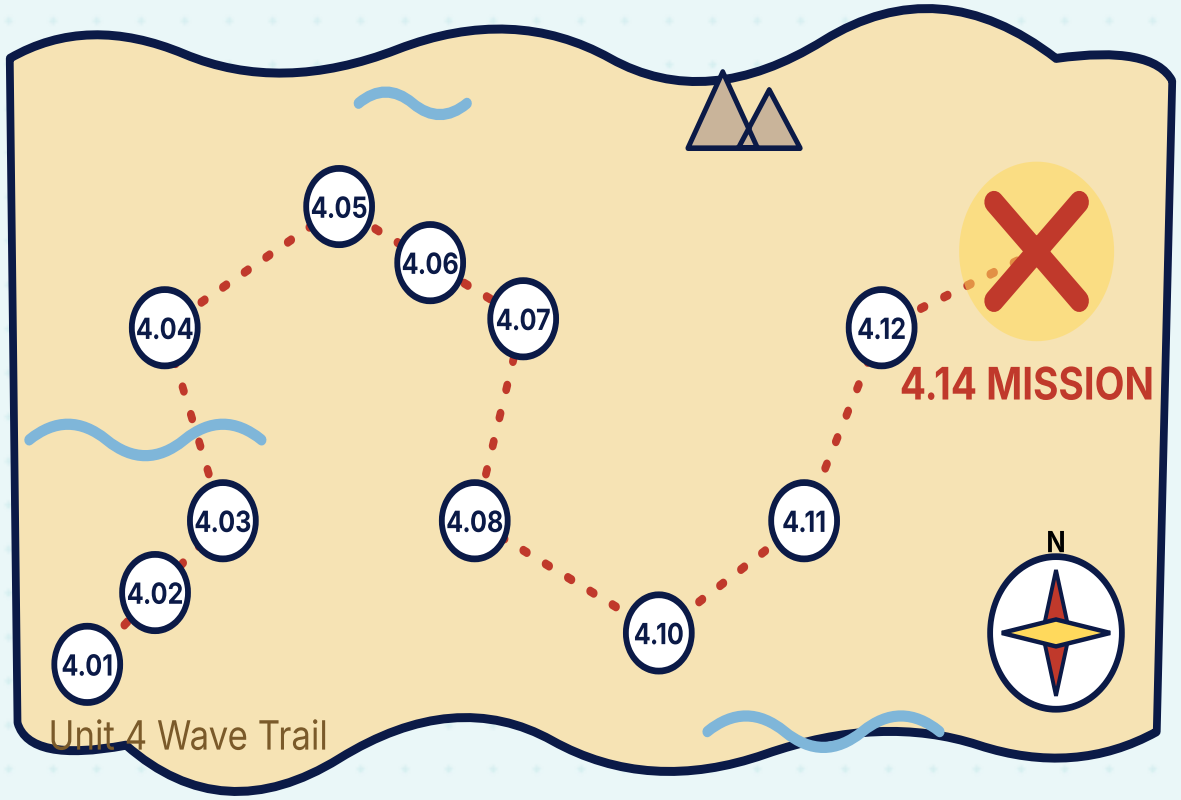
Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

Your Final Mission Awaits!

4.14 Unit 4 Mission · Waves Test

Every good adventurer takes a **map** on an important mission!





MISSION DUE
Mon, Dec 14

EXTENDED TIME
Mon, Dec 21

Don't wait until the trail goes cold! Finish by the due date.

The test is open book and open notes!

Pack your gear: your 4.14 Mission Map, Classkick notes, and Unit 4 vocabulary pages.

Friday is a school day! Spend at least 1 hour on science today.

Science Friday! Week 4 · Friday, Dec 4

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	CLASSKICK Power Up! + math Cases #1–#4	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.10, 4.11, 4.12, and 4.13 slides + Vocab Review	With 4.14 Test*	Yes / No
☐	CLASSKICK 4.14 Mission Map (done together on 4.13 day)	With 4.14 Test*	Yes / No
☐	SURVEY Unit 4 vocabulary post test	**Closes Thu, Dec 10**	Done = check it!

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished

Coming Up Next Week

Dec 7 – Dec 11 · Red tags show what **closes** that day.

MON 7	4.14 Unit 4 Mission (Test) ***NOTE: 4.14 Test assigned · due Mon, Dec 14 (ext Dec 21)***
TUE 8	Testing Day
WED 9	Testing Day
THU 10	Make-Up Day **CLOSES: UNIT 4 VOCAB POST TEST**
FRI 11	Science Friday!



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Science 7 · Week 5: Dec 7 – Dec 11

Unit 4: Wave Properties · Weekly Schedule

READ CLASSKICK ASSIGNED DUE!

MONDAY · DEC 7
4.14: Unit 4 Mission (Test)

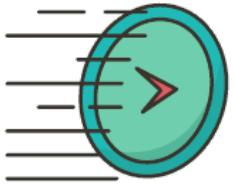
ASSIGNED **4.14 Test** (20 pts). Due Mon, Dec 14. *Extended time: Mon, Dec 21.*

GEAR Open book and open notes! Use your 4.14 Mission Map and Classkick notes.



TUESDAY · DEC 8
Testing Day

TESTING Follow your school testing schedule. No new science lesson.



WEDNESDAY · DEC 9
Testing Day

TESTING Follow your school testing schedule. No new science lesson.

CLASSKICK Extra time? Finish any Unit 4 Classkick slides.

THURSDAY · DEC 10
Make-Up Day

MAKE UP Make up any missing Unit 4 work.

DUE! **Unit 4 vocab post test** is due today.



FRIDAY · DEC 11
Science Friday

FINISH Finish any lessons, Classkick slides, or assignments you have not completed this week.

REDO Check your grades. Lower than a B? Review the Classkick notes and redo it.

CLASSKICK Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.

TEST The **4.14 Test** is due Monday, Dec 14. *Extended time: Mon, Dec 21.*

Friday is a school day! Spend at least 1 hour on science today.

Science Friday! Week 5 · Friday, Dec 11

My Assignment Checklist

1. Open your gradebook and find each assignment below.
2. Check the box when it is done and graded.
3. **Lower than a B?** Review the Classkick notes for that lesson, then **redo** the assignment.

Done	Assignment (open + assigned this week)	Closes	B or higher?
☐	GRADED **4.14 Test** (Unit 4 Mission) · 20 pts	**Closes Mon, Dec 14** ext: Mon, Dec 21	Yes No
☐	CLASSKICK **Full Unit 4 Classkick packet** (4.01–4.14 slides)	With 4.14 Test*	Yes No
☐	READ Any Unit 4 lesson you still need to read	No close date	Done = check it!

Coming Up Next Week

Dec 14 – Dec 18 · Red tags show what **closes** that day.

MON 14	Extended time · Make-up work · Math Case #1 **CLOSES: 4.14 TEST (REGULAR)** *DUE: Unit 4 Classkick packet*
TUE 15	Extended time · Make-up work · Math Case #2
WED 16	Extended time · Make-up work · Math Case #3
THU 17	Extended time · Make-up work · Math Case #4
FRI 18	Science Friday!

Class Padlet
Videos, slides, websites, and notes for every lesson

All done? Science Fun Padlet
Extra activities for when your work is finished



***Classkick reminder!** Your full Unit 4 Classkick packet is **due** with the 4.14 Unit Test (Mon, Dec 14; extended time Mon, Dec 21).

Science 7 · Week 6: Dec 14 – Dec 18

Unit 4: Wave Properties · Weekly Schedule

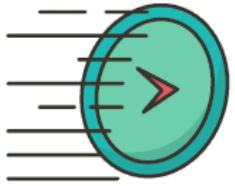
READ CLASSKICK ASSIGNED DUE!

MONDAY · DEC 14
Extended Time + Make-Ups

WARM-UP Integers: **Divide & Conquer!** + **Case #1** (cold front).

DUE! The **4.14 Test** and your full **Unit 4 Classkick packet** are due today (11:59 PM).

EXTENDED Extended time? Your test and packet close **Mon, Dec 21**.



TUESDAY · DEC 15
Extended Time + Make-Ups

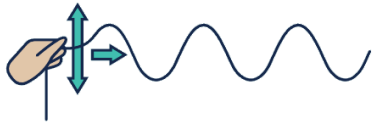
WARM-UP Integers: **Case #2** (lake-effect snow).

MAKE UP Extended time and make-up work. Finish any missing Unit 4 work.

WEDNESDAY · DEC 16
Extended Time + Make-Ups

WARM-UP Integers: **Case #3** (ash trees).

MAKE UP Extended time and make-up work. Finish any missing Unit 4 work.



THURSDAY · DEC 17
Extended Time + Make-Ups

WARM-UP Integers: **Case #4** (static charge).

MAKE UP Extended time and make-up work. Finish any missing Unit 4 work.

FRIDAY · DEC 18
Science Friday

FINISH Finish any lessons, Classkick slides, or assignments you have not completed this week.

REDO Check your grades. Lower than a B? Review the Classkick notes and redo it.

CLASSKICK Your full Unit 4 Classkick packet is due when the 4.14 Test is due and closes when the test closes.

BREAK Winter Break is Dec 21 – Jan 1. See you Mon, Jan 4!

INTEGER CREW: DIVIDE & CONQUER!

Dividing Integers

The crew must unlock the division gates. Divide the numbers, then choose the sign!

SAME signs? POSITIVE! | DIFFERENT signs? NEGATIVE!

SAME SIGNS → POSITIVE

Divide the counts. Make it positive.



$(-42) \div (-6) = 7$

$42 \div 6 = 7$

DIFFERENT SIGNS → NEGATIVE

Divide the counts. Make it negative.



$-35 \div 7 = -5$

$35 \div (-7) = -5$

CHECK IT: MULTIPLY BACK!

$-24 \div 4 = -6$ → $4 \times (-6) = -24$

Answer × divisor = starting number.

0 + a nonzero number = 0.
Never divide by 0.

Robots represent signed numbers.

CASE #1 • MONDAY

Is it cooling more than 5° an hour?

SCIENCE WORD: *Cold front*
Where cold, dense air pushes in under warmer air.

MYSTERY

A cold front moves across Ohio. Cold, dense air pushes in behind it and forces the warmer air up. Over 6 hours, the temperature in Columbus changes by −24°F. A TV weather reporter says it's dropping more than 5 degrees every hour. Is the reporter right?

CLUES

- Change: −24°F
- Time: 6 hours
- Reporter: over 5°

TEMP CHANGE (°F)





CLAIM

Is the weather reporter right?

☐ YES ☐ NO

EVIDENCE

Solve:

−24 ÷ 6 = _____

Check: _____ × 6 = −24

REASONING

Finish the sentence:

A negative divided by a positive is _____. It cooled _____ degrees an hour, which is _____ than 5.

CASE #2 • TUESDAY

Will Lake Erie freeze within a week?

SCIENCE WORD: *Lake-effect snow*

Snow made when cold air passes over warmer lake water.

MYSTERY

Lake-effect snow happens when cold air blows over the warmer, unfrozen water of Lake Erie. Water evaporates, then falls as snow on places like Cleveland and Chardon. During a cold snap, the lake must change by −18°F to start freezing, and it's changing by −2°F each day. Will it start freezing within 7 days?

CLUES

- Needed: −18°F
- Each day: −2°F
- 1 week = 7 days

LAKE CHANGE (°F)





CLAIM

Will the lake start freezing within 7 days?

☐ YES ☐ NO

EVIDENCE

Solve:

−18 ÷ (−2) = _____

Check: _____ × (−2) = −18

REASONING

Finish the sentence:

A negative divided by a negative is _____. It takes _____ days, which is _____ than 7 days.

CASE #3 • WEDNESDAY

Are the ash trees disappearing?

SCIENCE WORD: *Invasive species*
A living thing that spreads to a new place and harms it.

MYSTERY

The emerald ash borer is an invasive beetle that kills ash trees in Ohio forests. A park ranger compares tree counts. 3 years ago (−3 years), the park had 90 more ash trees than it has today (+90). Rate = change in trees ÷ change in time. Is the number of ash trees going down?

CLUES

- Trees then: +90
- Time: −3 years (the past)
- Now: 0

TREES VS. NOW





CLAIM

Is the number of ash trees going down?

☐ YES ☐ NO

EVIDENCE

Solve:

90 ÷ (−3) = _____

Check: _____ × (−3) = 90

REASONING

Finish the sentence:

A positive divided by a negative is _____. The park changes by _____ trees each year, so the number is going _____.

CASE #4 • THURSDAY

Does each sphere get a −9 charge?

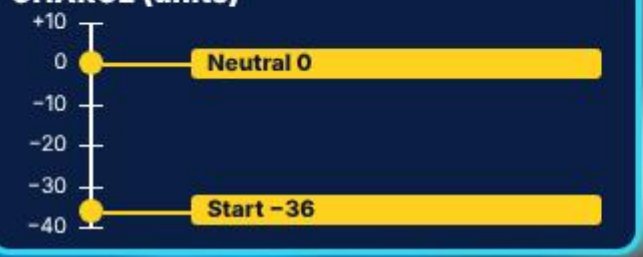
SCIENCE WORD: *Static charge*
A buildup of extra electric charge on an object.

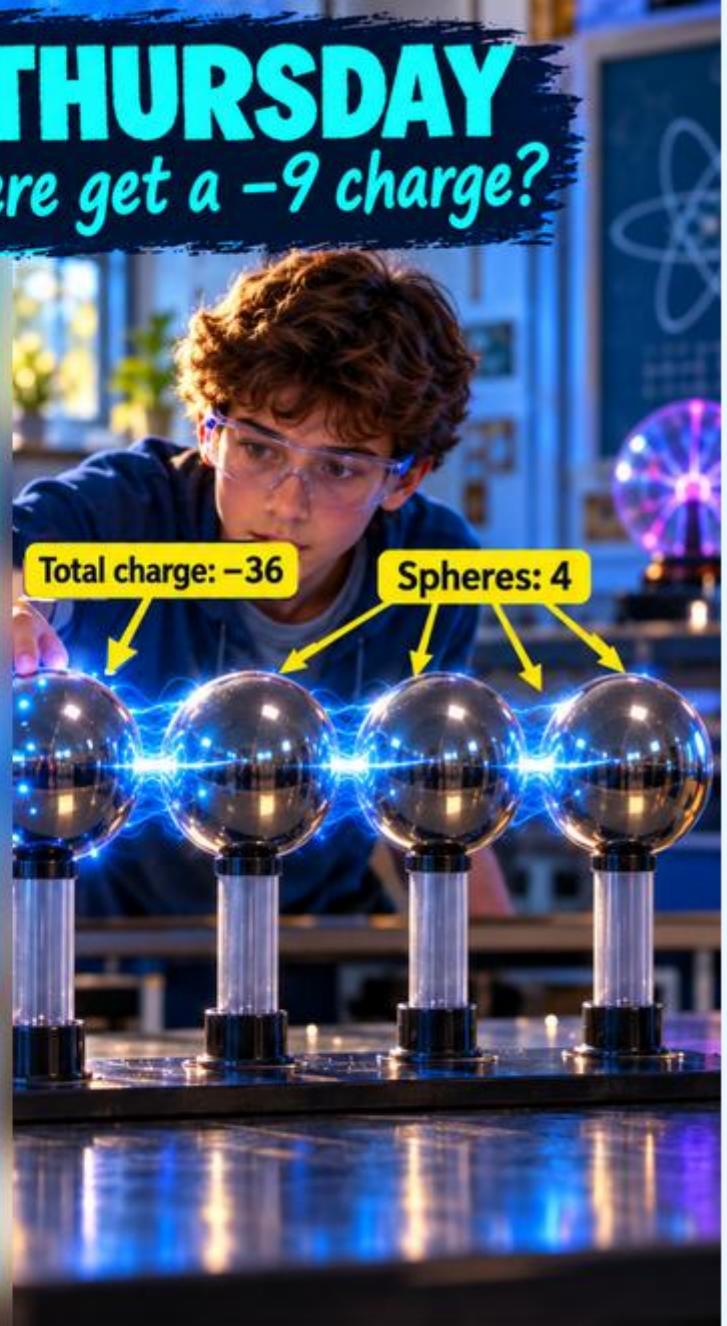
MYSTERY

In a static electricity demo, a metal sphere has extra electrons, giving it a charge of −36 units. It touches 3 identical metal spheres that have no charge. The extra electrons spread out evenly across all 4 spheres. Does each sphere end up with a charge of −9?

CLUES

- Total charge: −36
- Spheres: 4
- Shared evenly





CLAIM

Does each sphere end up with a charge of −9?

☐ YES ☐ NO

EVIDENCE

Solve:

$-36 \div 4 =$ _____

Check: _____ × 4 = −36

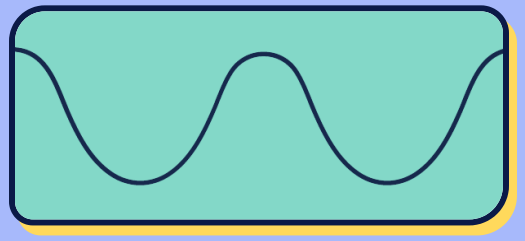
REASONING

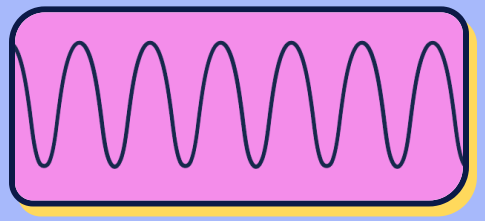
Finish the sentence:

A negative divided by a positive is _____. Each sphere gets a charge of _____ units.

This work hasn't been opened yet!







Check back soon!

Old Lessons (Not Used)

Saved for reference only

Old Lessons: 4.06 (old) and 4.07 (old)

These are the **old versions** of 4.06 and 4.07. They are **not part of this unit**. Do **not** fill them in unless your teacher tells you to. The new 4.06 (Transverse Waves) and 4.07 (Longitudinal Waves) are earlier in your packet.

Teacher note: kept for reference: Mechanical vs. EM, Sound vs. Light, How Sound Travels lab, Transverse and Longitudinal, Sound Wave Diagram.

4.06 (old): Types of Waves

Part 1: Mechanical vs. Electromagnetic Waves

Interactive Link
Mechanical vs. Electromagnetic Waves Video (4.06 page)

★ Key Ideas

- **Mechanical wave**: moves **energy** through a **medium** by vibrating its particles. Examples: sound, water waves.
- **Electromagnetic (EM) wave**: does **not** need a medium. It can travel through empty space. Examples: light, radio, X-rays.
- **Transverse**: the medium moves **up and down**, at a right angle to the wave.
- **Longitudinal:** the medium moves **back and forth**, in the same direction as the wave.

boxed = Unit 4 focus word

1 Drag each card into the correct column. Use each card once. DRAG

radio waves | sound | X-rays | earthquake waves | visible light | water waves | microwaves | ultrasound

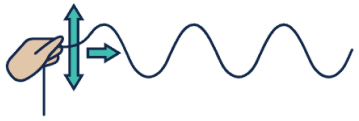
Mechanical	Electromagnetic
needs a medium	no medium needed

2 Watch the video on the 4.06 page. Then circle your answers. CIRCLE

USE THE LINK ON THIS SLIDE

	Mechanical		EM	
Needs a medium (solid, liquid, gas)?	YES	NO	YES	NO
Can travel through empty space?	YES	NO	YES	NO
Has wavelength, frequency, amplitude, and speed?	YES	NO	YES	NO

3 Drag a word into each blank. DRAG





In a transverse wave, the medium moves _____. In a longitudinal wave, the medium moves _____. A wave that needs a medium is a _____ wave.

mechanical | back and forth | longitudinal | electromagnetic | up and down | transverse | *One extra word.*

4 The Sun is very loud, but we cannot hear it. We can see its light. Drag a word into each blank. DRAG

Space is almost empty, so there is no _____. Sound is a _____ wave, so it cannot reach us. Light is an _____ wave, so it can travel through _____.

medium | electromagnetic | empty space | transverse | mechanical | *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.06 (old): Types of Waves

Part 2: Sound Waves vs. Light Waves

Interactive Link
Sound vs. Light Sort (Virtual Science Teachers)

★ Key Ideas

- **Sound** is a **mechanical**, longitudinal **wave**. It needs a **medium**.
- **Light** is an **electromagnetic** wave. It can travel through empty space.
- Both carry **energy** and have **frequency**, **wavelength**, and **amplitude**.
- Sound frequency = **pitch**. Light wavelength = **color**.
- Light is much faster, so we see lightning before we hear thunder.

boxed = Unit 4 focus word

5 Open the **Sound vs. Light Sort**. Drag each card and press **Check**. Then circle **Sound**, **Light**, or **Both**. CIRCLE USE THE LINK ON THIS SLIDE

Needs a medium (solid, liquid, or gas)	Sound	Light	Both	Can travel through empty space	Sound	Light	Both
Carries energy	Sound	Light	Both	Is a mechanical wave	Sound	Light	Both
Is an electromagnetic wave	Sound	Light	Both	Has frequency, wavelength, and amplitude	Sound	Light	Both
Travels about 340 m/s in air	Sound	Light	Both	Travels about 300,000 km/s	Sound	Light	Both

6 Drag a word into each blank. DRAG

Different frequencies of sound are heard as different _____. Different wavelengths of light are seen as different _____. Sound is a _____ wave, and light is an _____ wave.

pitches | electromagnetic | mechanical | colors | speeds

One extra word.

7 Lightning and thunder happen at the same time. Circle your answers. CIRCLE

a) Which reaches you first? the lightning (light) | the thunder (sound)

b) Why? Light travels faster | slower than sound.

c) Sound goes about 340 m each second. You count 3 seconds before the thunder. The storm is about 100 m | 1 km | 10 km away.

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.06 (old): Types of Waves

Lab: How Sound Travels (Fred the Frog)

Interactive Link 1
Animated Dominoes Video

Interactive Link 2
How Sound Travels (NASA)

★ Key Ideas

- Sound is a **mechanical** **wave**. It needs particles to pass the **energy** along.
- Sound travels **fastest in solids**, faster in liquids, and **slowest in gases**.
- There are no particles in space, so **sound cannot travel in space**.
- Light and radio are **EM waves**. They travel through space, and they are much faster than sound.

boxed = Unit 4 focus word

P **Predict! Watch the domino video (Link 1). The dominoes are like particles.** CIRCLE

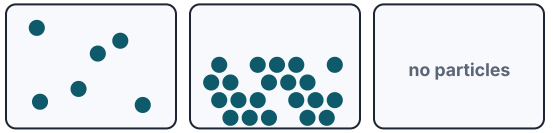
Close-together dominoes are like a (solid) (gas). Far-apart dominoes are like a (solid) (gas)

I predict sound will travel fastest through (air) (water) (outer space)

Lab Steps

- Open the NASA lab (Link 2). Read the first page about **gas, liquid, and solid**. Click **Next**.
- Choose a place: **Land** (air), **Water**, or **Space**.
- Click **Croak**. Fred the Frog is 200 meters from the man.
- Wait for the timer to stop. Record the time. Try all three places.
- Read the red-dot buttons at the top for help.

Think: How close together are the particles in each medium? Was your prediction right?



gas (air) | liquid (water) | space (empty)

8 **Record your data.** FILL IN USE THE LINK ON THIS SLIDE

Medium	Particles: really far apart, close together, or none?	Time to travel 200 m (s)
Gas (Land)		
Liquid (Water)		
Space		

9 **Use your data. Circle your answers.** CIRCLE

a) Sound was fastest in (air) (water) (space)

b) In space, Fred's croak (was heard) (was never heard)

c) Which is faster? (light (EM)) (sound (mechanical))

d) Astronauts talk by radio. A radio wave is (mechanical) (EM)

10 **Why did the croak travel faster in water, and not at all in space? Drag a word into each blank.** DRAG

Water particles are _____ than air particles, so they pass the vibration along _____. Space has _____, so sound cannot travel there.

(no particles) (closer together) (farther apart) (faster) *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.07 (old): Mechanical Waves

Transverse and Longitudinal Waves

Interactive Link 1
Make a Wave (PBS)

Interactive Link 2
Up & Down? Side to Side? (SERP)

★ Key Ideas

- **Transverse wave**: the **medium** vibrates at **right angles (perpendicular)** to the direction the wave travels.
- **Longitudinal wave:** the medium vibrates **parallel** to the direction the wave travels.
- **Compression:** particles crowd together. **Rarefaction:** particles spread apart.
- More **energy** = tighter compressions and wider rarefactions.

boxed = Unit 4 focus word

1 Open Link 1. Click each button and watch the coils of the spring. CIRCLE USE THE LINK ON THIS SLIDE

⇅ **button:** the coils move [up and down] [back and forth]

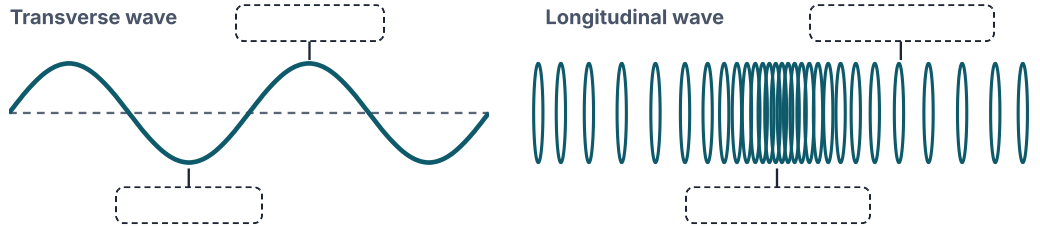
This makes a [transverse] [longitudinal] wave.

⇄ **button:** the coils move [up and down] [back and forth]

This makes a [transverse] [longitudinal] wave.

After the wave passes, the coils
[return to where they started] [move to the end]

2 Drag a word into each box. DRAG



rarefaction | crest | normal | compression | trough *One extra word.*

3 Try both wave models in Link 2. Then circle the type of wave. CIRCLE USE THE LINK ON THIS SLIDE

A guitar string is plucked.	transverse	longitudinal
A slinky is pushed in and out.	transverse	longitudinal
A rope is shaken up and down.	transverse	longitudinal
A sound wave moves through air.	transverse	longitudinal

In a longitudinal wave, the energy and the disturbance move
[parallel] [perpendicular] to each other.

4 You push a slinky gently, then push it hard. Drag a word into each blank. DRAG

The hard push gives the wave more _____. The coils in the compressions get _____, and the rarefactions spread _____. The wave has a bigger _____.

farther apart | frequency | energy | closer together | amplitude *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.

4.07 (old): Mechanical Waves

Part 2: Sound Wave Diagram

Interactive Link
Sound Wave Diagram (Virtual Science Teachers)

★ Key Ideas

- Sound is a **longitudinal** **wave**, but we often draw it as a **transverse**-looking graph.
- On the graph, a **crest** shows a **compression** and a **trough** shows a **rarefaction**.
- **Wavelength** = compression to compression (or crest to crest).
- More **energy** = more crowded compressions = taller crests (bigger **amplitude**).

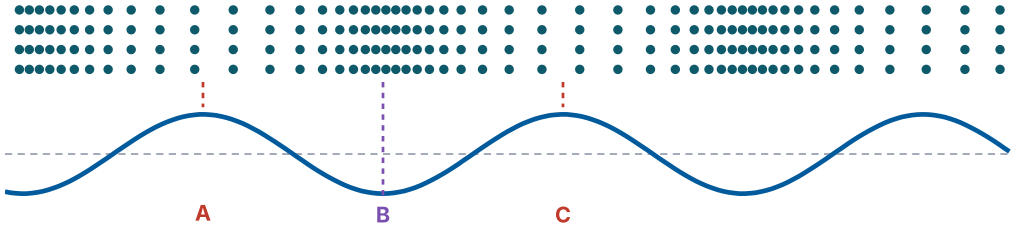
boxed = Unit 4 focus word

5 Open the **Sound Wave Diagram**. Drag each label and press **Check**. Then drag a word into each box. DRAG USE THE LINK ON THIS SLIDE

Definition	Word
The highest point of a transverse wave	
The lowest point of a transverse wave	
Particles squeezed close together	
Particles spread farther apart	
Distance between repeating points on a wave	
Height of a wave from its rest position	

amplitude | frequency | wavelength | compression | crest | rarefaction | trough *One extra word.*

6 The particle model and the graph show the same sound wave. Circle your answers. CIRCLE



a) At A and C, the particles are in a **compression** **rarefaction**. On the graph, that is a **crest** **trough**.

b) At B, the particles are in a **compression** **rarefaction**. On the graph, that is a **crest** **trough**.

c) One wavelength is from **A to B** **A to C**

7 A sound gets louder. Drag a word into each blank. DRAG

A louder sound has more _____. The particles in the compressions are packed _____, so the crests on the graph are _____.

taller | closer together | energy | farther apart *One extra word.*

Reminder: Your full Unit 4 Classkick packet is due when the Unit 4 test is due, and it closes when the test closes.